MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

ENCODER, PRIORITY, KEY CODE NTA

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic priority encoder module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.12174 failures per million operating hours for class I or 0.30047 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be NTA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/338-1 for class I and M28787/338-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

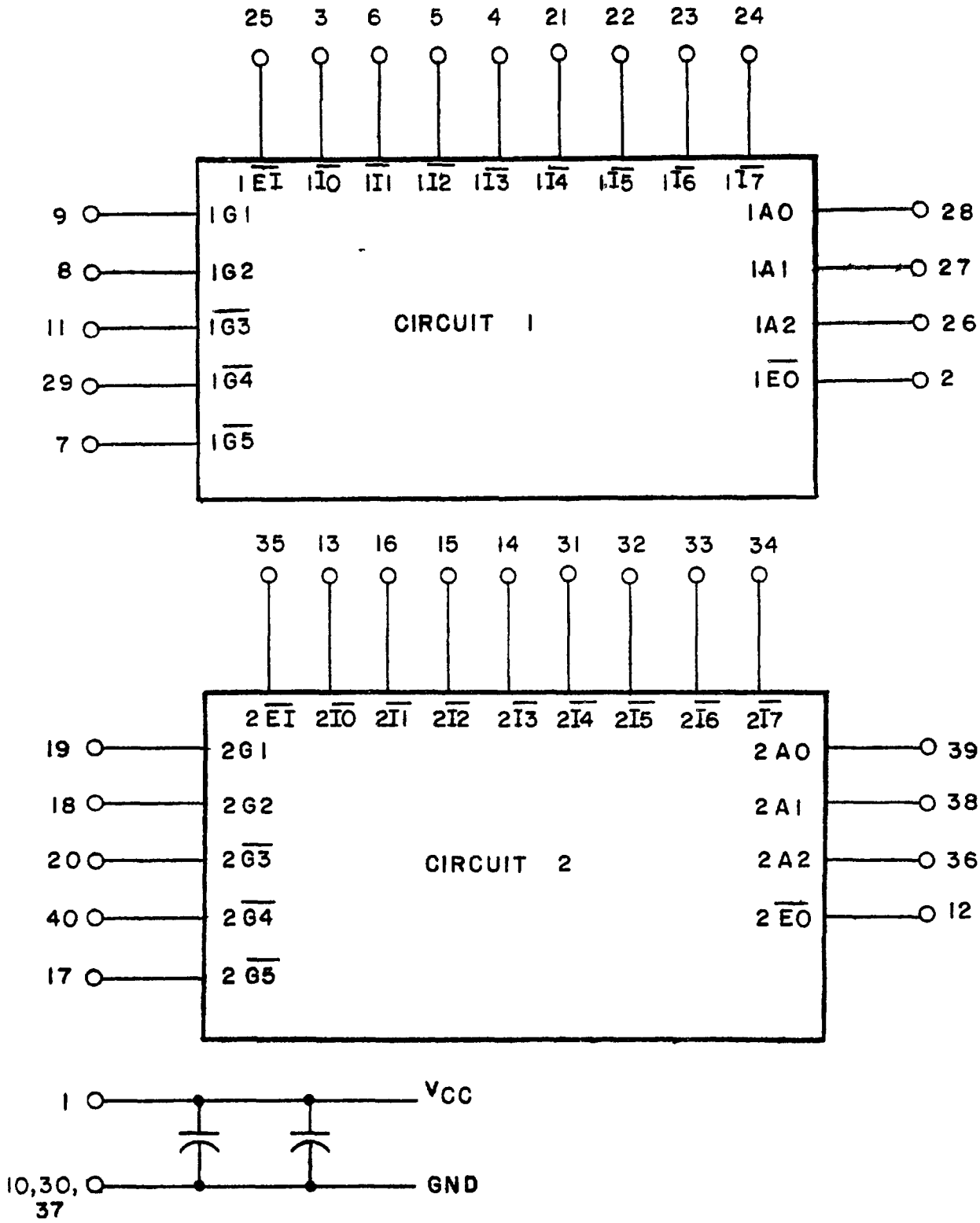


FIGURE 1. NTA module function diagram, priority encoder.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	IV	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	IV	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1,B2	Continuity, ground pin	V	X	0	1	0	1	0	1	0	1	0	1	Ω
C	V_{CC} voltage stress	VI	S	← Module shall pass all remaining tests →										---
D1-D14	Input breakdown current	VII	S	0	100	0	100	0	100	0	100	0	100	µA
D15-D28	Input breakdown current	VII	S	0	200	0	200	0	200	0	200	0	200	µA
E1-E28	Input clamp diode voltage	VIII	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
F1-F8	Output short circuit current	IX	S	-15	-85	-15	-85	-15	-85	-15	-85	-15	-85	mA
G1-G18	Functional	X	X	← See table X →										---
H1-H14	Input current, high level	XI	X	0	20	0	20	0	20	0	20	0	20	µA
H15-H28	Input current, high level	XI	X	0	40	0	40	0	40	0	40	0	40	µA
J1-J14	Input current, low level	XII	X	-10	-440	-10	-440	-10	-440	-10	-440	-10	-440	µA
J15-J28	Input current, low level	XII	X	-10	-870	-10	-870	-10	-870	-10	-870	-10	-870	µA
K1,K2	Output voltage, high level	XIII	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
K3-K8	Output voltage, high level	XIII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
L1,L2	Output voltage, low level	XIV	X	0	400	0	400	0	400	0	400	0	400	mV
L3-L8	Output voltage, low level	XIV	X	0	500	0	500	0	500	0	500	0	500	mV
M1-M6	High impedance leakage current, high level	XV	X	-1	20	-1	20	-1	20	-1	20	-1	20	µA

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N1-N6	High impedance leakage current, low level	XV	X	-20	1	-20	1	-20	1	-20	1	-20	1	µA
P	Power supply current	XVI	X	5	48	5	48	5	48	5	48	5	48	mA
Q1-Q16	Propagation delay time (t_{PLH}), out-of-phase, C_L = 50 pF	XVII	S	3	15	3	16	3	16	3	18	3	18	ns
Q17,Q18	Propagation delay time (t_{PLH}), inphase, C_L = 50 pF	XVII	S	5	21	5	22	5	22	5	24	5	24	ns
Q19-Q24	Propagation delay time (t_{PLH}), out-of-phase, C_L = 50 pF	XVII	S	6	22	6	23	6	23	6	25	6	25	ns
Q25-Q48	Propagation delay time (t_{PLH}), out-of-phase, C_L = 50 pF	XVII	S	5	21	5	24	5	24	5	27	5	27	ns
Q49-Q58	Propagation delay time (t_{PLH}), inphase, C_L = 50 pF	XVII	S	8	29	8	33	8	33	8	37	8	37	ns
R1-R16	Propagation delay time (t_{PHL}), out-of-phase, C_L = 50 pF	XVII	S	12	40	12	50	12	50	12	60	12	60	ns
R17,R18	Propagation delay time (t_{PHL}), inphase, C_L = 50 pF	XVII	S	11	38	11	47	11	47	11	57	11	57	ns
R19-R24	Propagation delay time (t_{PHL}), out-of-phase, C_L = 50 pF	XVII	S	7	25	7	28	7	28	7	32	7	32	ns
R25-R48	Propagation delay time (t_{PHL}), out-of-phase, C_L = 50 pF	XVII	S	6	22	6	23	6	23	6	25	6	25	ns
R49-R58	Propagation delay time (t_{PHL}), inphase, C_L = 50 pF	XVII	S	8	29	8	31	8	31	8	34	8	34	ns
S1-S16	Propagation delay time (t_{PLH}), out-of-phase, C_L = 150 pF	XVII	Q	3	25	3	26	3	26	3	28	3	28	ns

See footnotes at end of table.

Source: https://assist.dla.mil - Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S17,S18	Propagation delay time (t_{PLH}), inphase, C_L = 150 pF	XVII	Q	5	31	5	32	5	32	5	34	5	34	ns
S19-S24	Propagation delay time (t_{PLH}), out-of-phase, C_L = 150 pF	XVII	Q	6	32	6	33	6	33	6	35	6	35	ns
S25-S48	Propagation delay time (t_{PLH}), out-of-phase, C_L = 150 pF	XVII	Q	6	31	6	34	6	34	6	37	6	37	ns
S49-S58	Propagation delay time (t_{PLH}), inphase, C_L = 150 pF	XVII	Q	8	39	8	43	8	43	8	47	8	47	ns
T1-T16	Propagation delay time (t_{PHL}), out-of-phase, C_L = 150 pF	XVII	Q	12	50	12	60	12	60	12	70	12	70	ns
T17,T18	Propagation delay time (t_{PHL}), inphase, C_L = 150 pF	XVII	Q	11	48	11	57	11	57	11	67	11	67	ns
T19-T24	Propagation delay time (t_{PHL}), out-of-phase, C_L = 150 pF	XVII	Q	7	35	7	38	7	38	7	42	7	42	ns
T25-T48	Propagation delay time (t_{PHL}), out-of-phase, C_L = 150 pF	XVII	Q	6	32	6	33	6	33	6	35	6	35	ns
T49-T58	Propagation delay time (t_{PHL}), inphase, C_L = 150 pF	XVII	Q	8	39	8	41	8	41	8	44	8	44	ns
U1-U12	Disable recovery time, t_{PHZ}	XVIII	S	8	60	8	60	8	60	8	60	8	60	ns
U13-U30	Disable recovery time, t_{PHZ}	XVIII	S	8	55	8	55	8	55	8	55	8	55	ns
U31-U42	Disable recovery time, t_{PLZ}	XVIII	S	9	60	9	60	9	60	9	60	9	60	ns
U43-U60	Disable recovery time, t_{PLZ}	XVIII	S	9	55	9	55	9	55	9	55	9	55	ns
V1-V12	Enable recovery time, t_{PZH}	XIX	S	11	49	11	49	11	54	11	54	11	54	ns
V13-V30	Enable recovery time, t_{PZH}	XIX	S	10	43	10	43	10	47	10	47	10	47	ns

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
V31-V42	Enable recovery time, t_{PZL}	XIX	S	10	49	10	49	10	49	10	49	10	49	ns
V43-V60	Enable recovery time, t_{PZL}	XIX	S	9	43	9	43	9	43	9	43	9	43	ns
W1-W6	Output crosstalk, low level	XX	Q	0	400	0	400	0	400	0	400	0	400	mV
X1-X8	Output crosstalk, high level	XX	Q	0	400	0	400	0	400	0	400	0	400	mV
Y	Power supply transient amplitude	XXI	Q	0	400	0	400	0	400	0	400	0	400	mV
Z	Power supply filter capacitance	XXII	S	1.54	2.86	1.54	2.86	1.54	2.86	1.54	2.86	1.54	2.86	µF
AA1-AA14	Input capacitance	XXIII	Q	2	15	2	15	2	15	2	15	2	15	pF
AA15-AA28	Input capacitance	XXIII	Q	2	20	2	20	2	20	2	20	2	20	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through AA shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through AA in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions G through Z shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition Z, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests G through Z shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests G through Z shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation	264	264	mW
Input voltage			
Low level	800 max	800 max	mV
High level	2 min	2 min	V
Input current			
Low level (V_{IL} = 0 V)			
$\overline{EI}$, G1, G2, $\overline{G3}$, $\overline{G4}$, $\overline{G5}$, $\overline{IO}$	-10 min, -440 max	-10 min, -440 max	µA
All others	-10 min, -870 max	-10 min, -870 max	µA
High level			
V_{IH} = 2.7 V ($\overline{EI}$, G1, G2, $\overline{G3}$, $\overline{G4}$, $\overline{G5}$, $\overline{IO}$)	0 min, 20 max	0 min, 20 max	µA
(All others)	0 min, 40 max	0 min, 40 max	µA
V_{IH} = 5.5 V ($\overline{EI}$, G1, G2, $\overline{G3}$, $\overline{G4}$, $\overline{G5}$, $\overline{IO}$)	0 min, 100 max	0 min, 100 max	µA
(All others)	0 min, 200 max	0 min, 200 max	µA
Output drive current			
Low level			
V_{OL} = 400 mV ($\overline{EO}$)	4 max	4 max	mA
V_{OL} = 500 mV (All others)	12 max	12 max	mA
High level			
V_{OH} = 2.5 V ($\overline{EO}$)	-440 max	-440 max	µA
V_{OH} = 2.4 V (All others)	-1 max	-1 max	mA

6.5 Module functions This module contains two completely independent 8-line to 3-line binary priority encoders. The circuits accept active low inputs and produce a three-state active high output prioritized from active $\overline{I7}$ to $\overline{I0}$. The output has five control signals, three active low and two active high. Each circuit has a cascade input and enable output.

6.5.1 Input to output relationship. Each priority encoder has the input to output relationship of tables II and III for positive logic.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

Agent:
DLA - ES

Preparing Activity:
Navy - EC

(Project 5963-N636)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE II. Input to output relationship. 1/

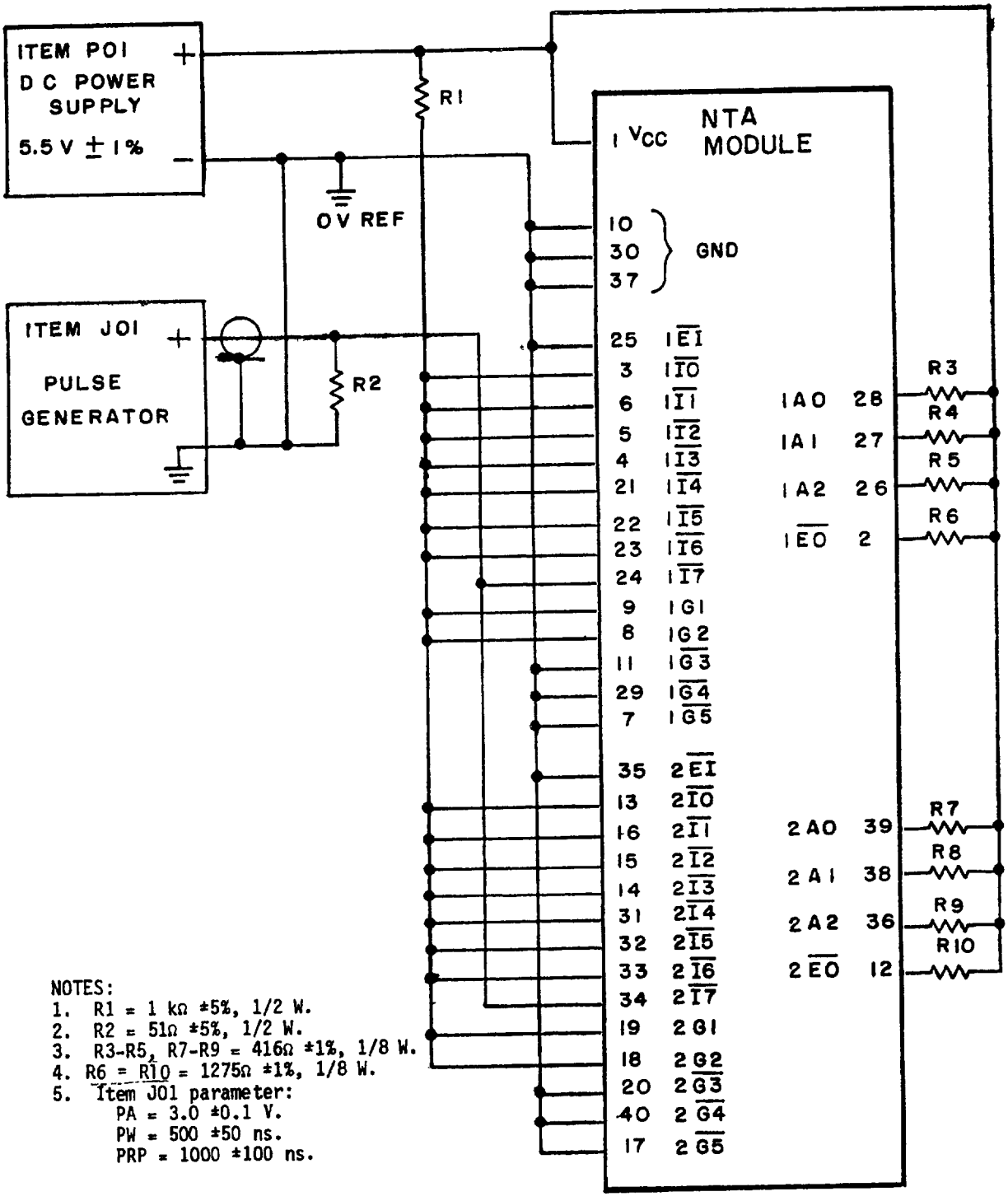
Inputs									Outputs			
$\overline{EI}$	$\overline{I0}$	$\overline{I1}$	$\overline{I2}$	$\overline{I3}$	$\overline{I4}$	$\overline{I5}$	$\overline{I6}$	$\overline{I7}$	A0	A1	A2	$\overline{EO}$
H	X	X	X	X	X	X	X	X	L	L	L	H
L	H	H	H	H	H	H	H	H	L	L	L	L
L	X	X	X	X	X	X	X	L	H	H	H	H
L	X	X	X	X	X	X	L	H	L	H	H	H
L	X	X	X	X	X	L	H	H	H	L	H	H
L	X	X	X	X	L	H	H	H	L	L	H	H
L	X	X	X	L	H	H	H	H	H	H	L	H
L	X	X	L	H	H	H	H	H	L	H	L	H
L	X	L	H	H	H	H	H	H	H	L	L	H
L	L	H	H	H	H	H	H	H	L	L	L	H

TABLE III. Input to output relationship. 1/

G1	G2	$\overline{G3}$	$\overline{G4}$	$\overline{G5}$	A0	A1	A2
H	H	L	L	L	Enabled		
L	X	X	X	X	Z	Z	Z
X	L	X	X	X	Z	Z	Z
X	X	H	X	X	Z	Z	Z
X	X	X	H	X	Z	Z	Z
X	X	X	X	H	Z	Z	Z

1/ H = High level
L = Low level
X = Don't care
Z = High impedance

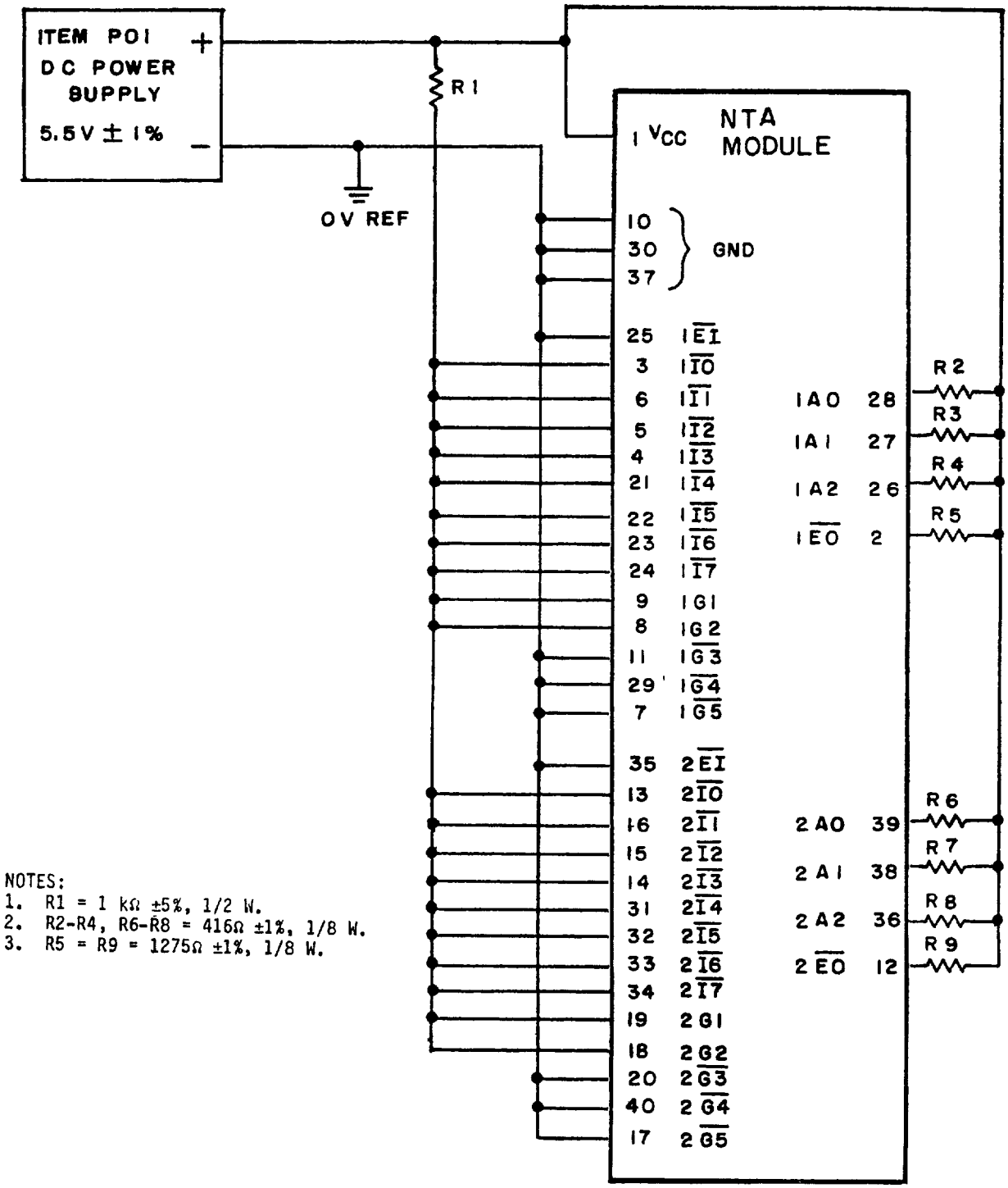
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 1 kΩ ±5%, 1/2 W.
2. R2 = 51Ω ±5%, 1/2 W.
3. R3-R5, R7-R9 = 416Ω ±1%, 1/8 W.
4. R6 = R10 = 1275Ω ±1%, 1/8 W.
5. Item J01 parameter:
   PA = 3.0 ±0.1 V.
   PW = 500 ±50 ns.
   PRP = 1000 ±100 ns.

FIGURE 2. Operating temperature and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 1 kΩ ±5%, 1/2 W.
2. R2-R4, R6-R8 = 416Ω ±1%, 1/8 W.
3. R5 = R9 = 1275Ω ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

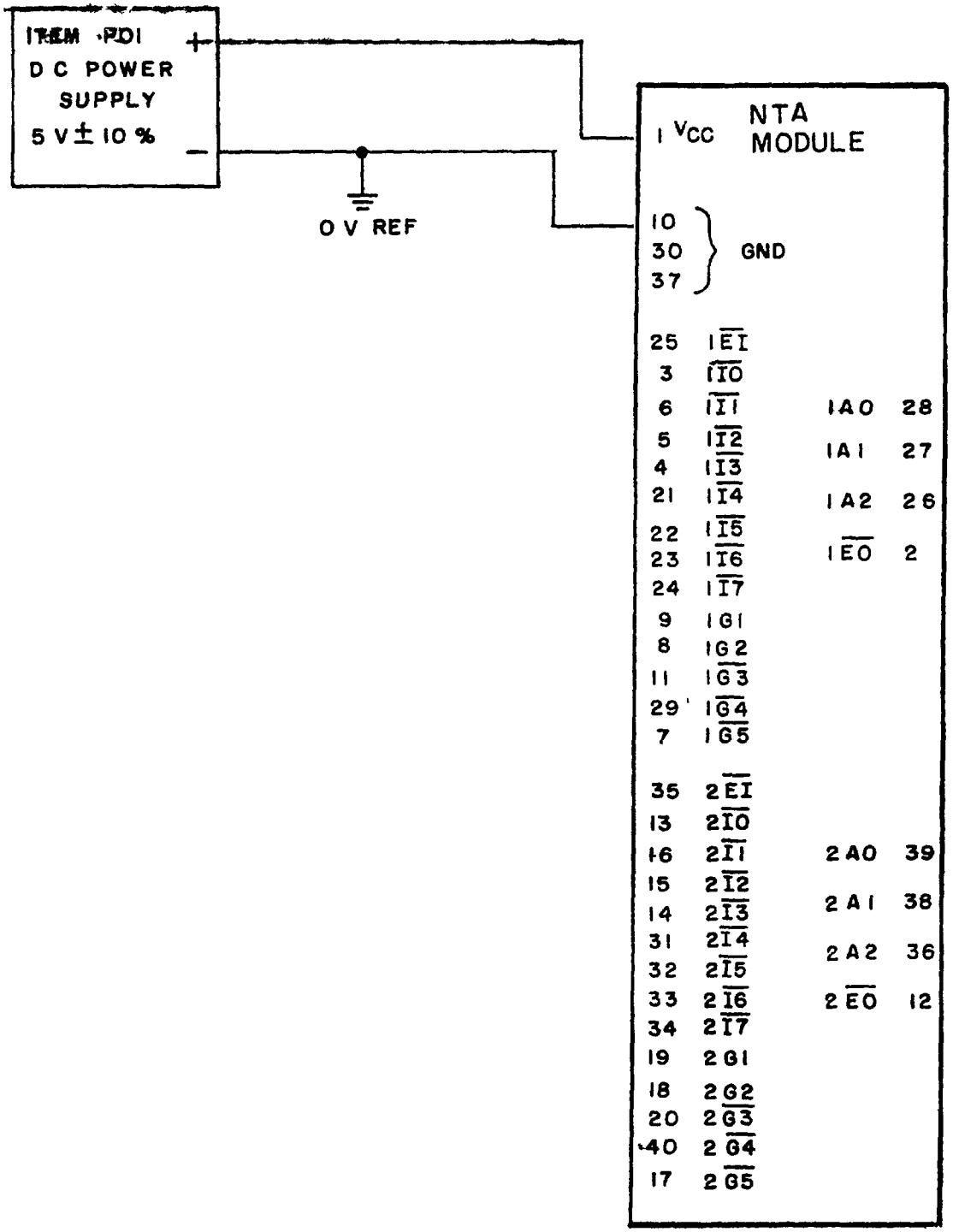


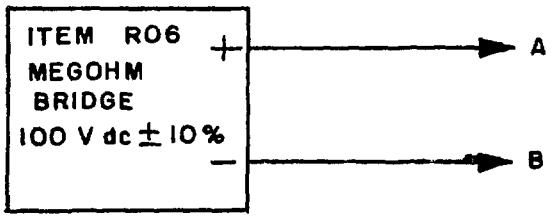
FIGURE 4. Vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE IV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation, frame	All pins	Frame	10	∞	MΩ	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

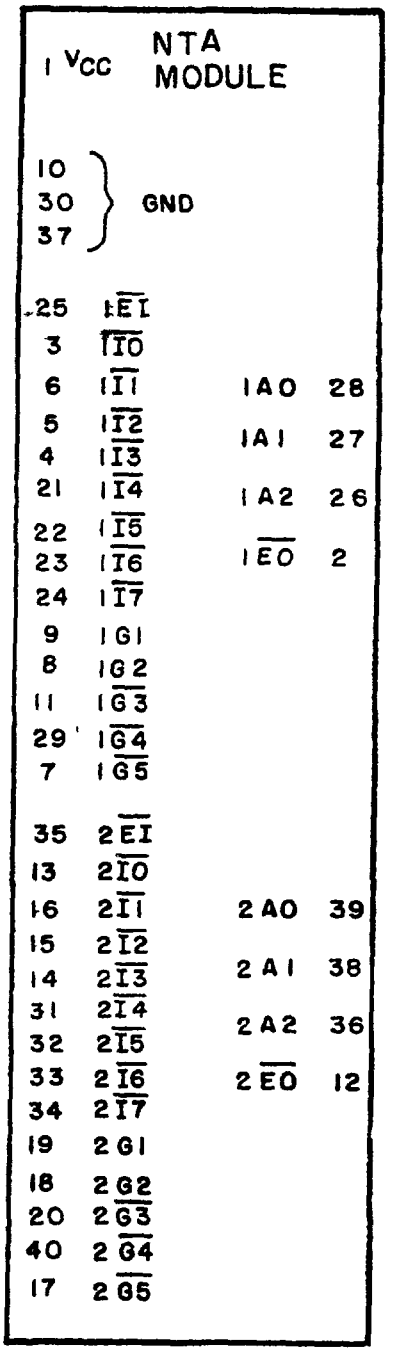


FIGURE 5. Isolation test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE V. Continuity (ground pin) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Continuity, ground pin	10	30	0	1	Ω	6
B2	Continuity, ground pin	10	37	0	1	Ω	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

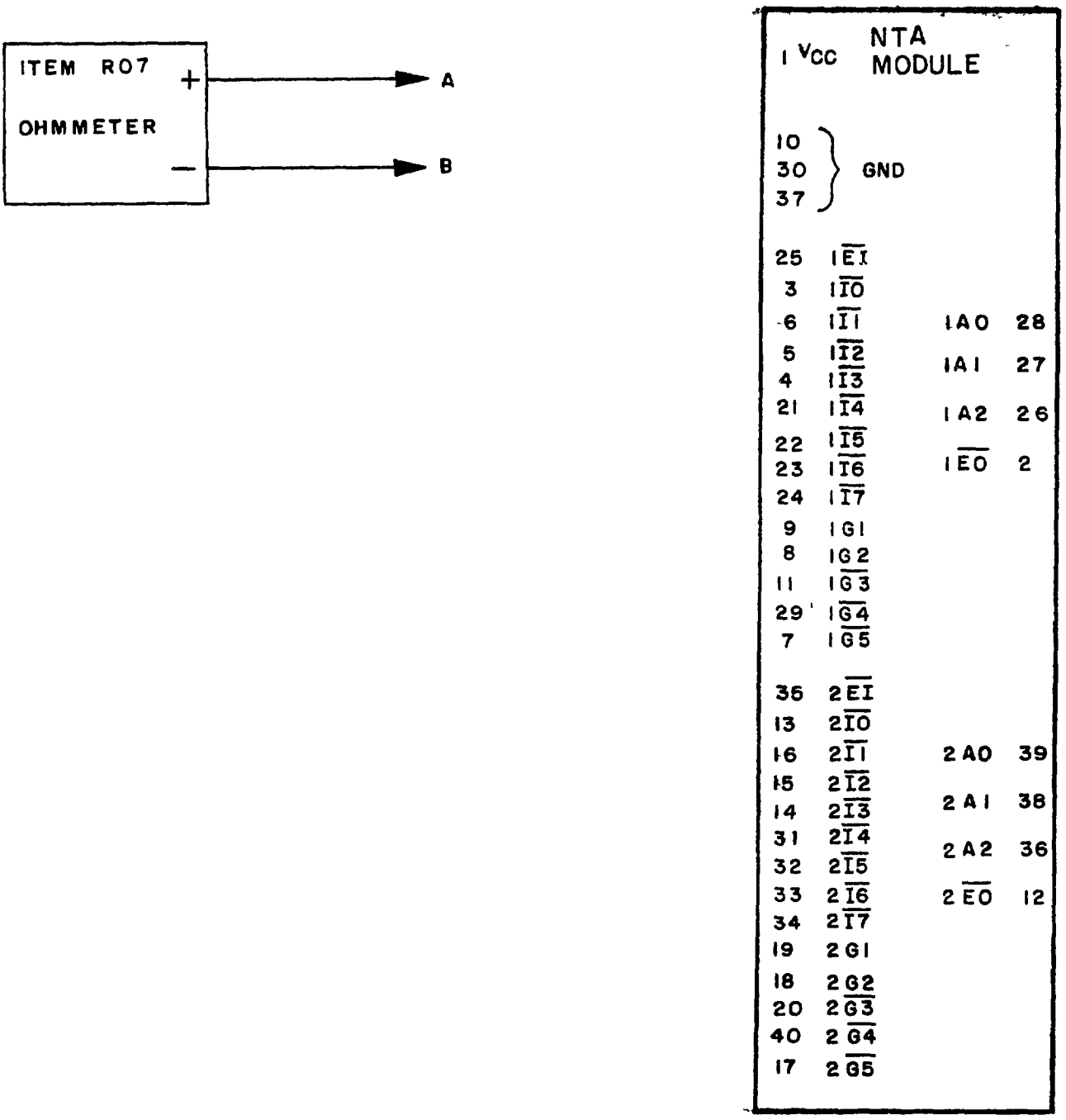


FIGURE 6. Continuity (ground pin) test circuit for table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE VI. V_{CC} voltage stress test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
C	V_{CC} voltage stress	See table I		---	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

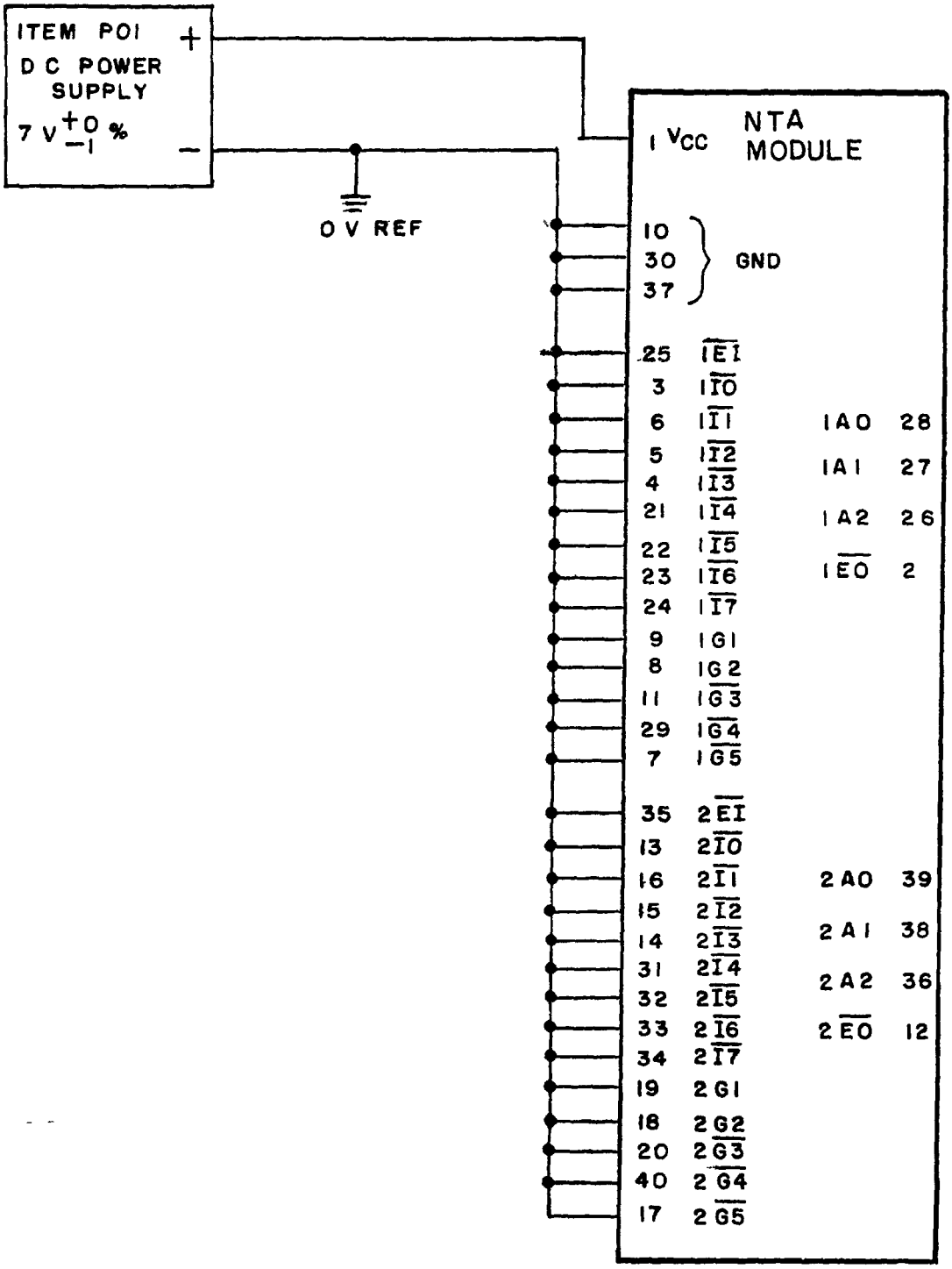


FIGURE 7. V_{CC} voltage stress test circuit for table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE VII. Input breakdown current test.

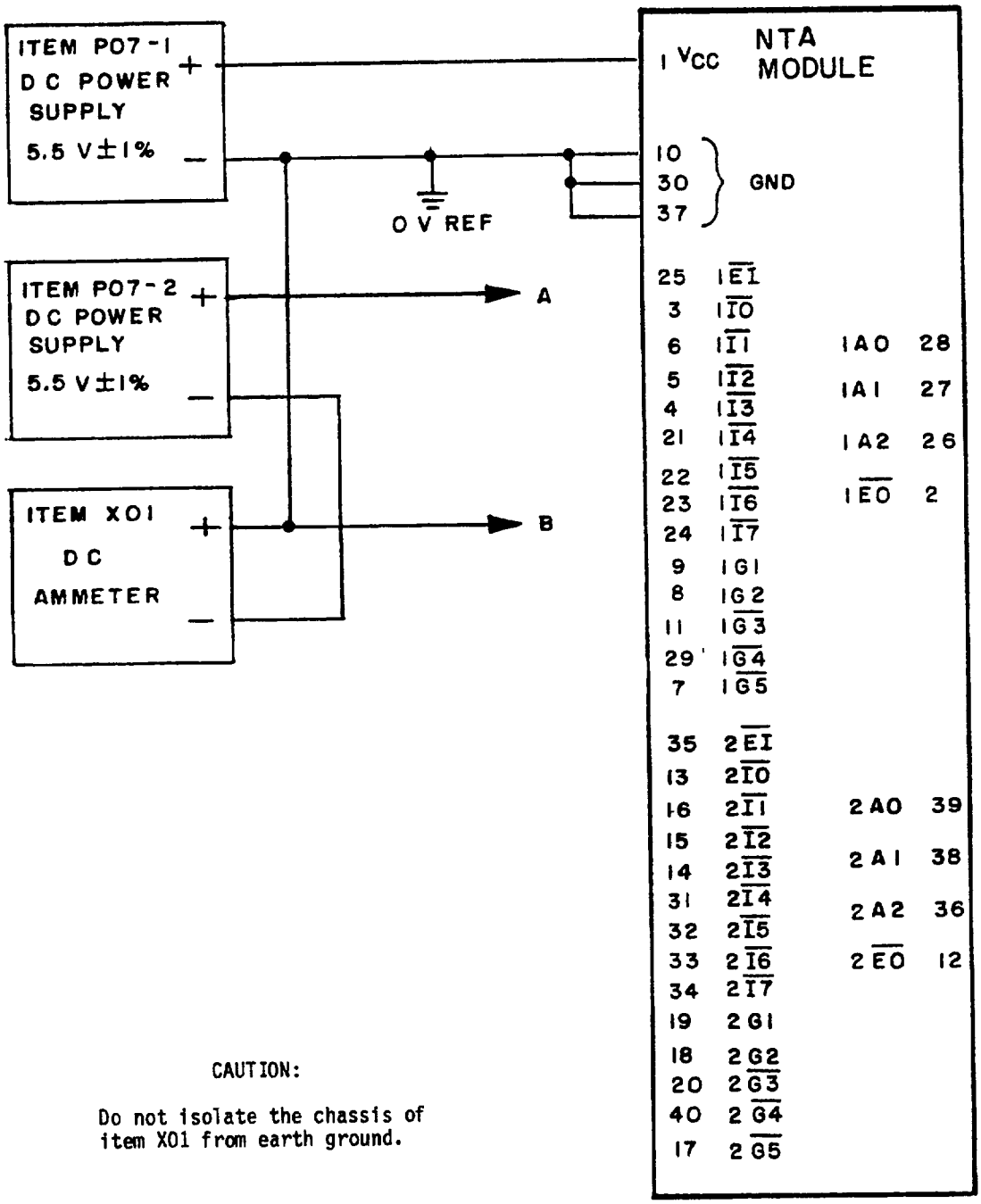
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	$1\overline{EI}$ Input breakdown current	25	All input pins except pin under test	0	100	µA	8
D2	$1\overline{I0}$ "	3	"	"	"	"	"
D3	1G1 "	9	"	"	"	"	"
D4	1G2 "	8	"	"	"	"	"
D5	$1\overline{G3}$ "	11	"	"	"	"	"
D6	$1\overline{G4}$ "	29	"	"	"	"	"
D7	$1\overline{G5}$ "	7	"	"	"	"	"
D8	$2\overline{EI}$ "	35	"	"	"	"	"
D9	$2\overline{I0}$ "	13	"	"	"	"	"
D10	2G1 "	19	"	"	"	"	"
D11	2G2 "	18	"	"	"	"	"
D12	$2\overline{G3}$ "	20	"	"	"	"	"
D13	$2\overline{G4}$ "	40	"	"	"	"	"
D14	$2\overline{G5}$ "	17	"	"	"	"	"
D15	$1\overline{I1}$ "	6	"	"	200	"	"
D16	$1\overline{I2}$ "	5	"	"	"	"	"
D17	$1\overline{I3}$ "	4	"	"	"	"	"
D18	$1\overline{I4}$ "	21	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE VII. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D19	$1\overline{I5}$ Input breakdown current	22	All input pins except pin under test	0	200	µA	8
D20	$1\overline{I6}$ "	23	"	"	"	"	"
D21	$1\overline{I7}$ "	24	"	"	"	"	"
D22	$2\overline{I1}$ "	16	"	"	"	"	"
D23	$2\overline{I2}$ "	15	"	"	"	"	"
D24	$2\overline{I3}$ "	14	"	"	"	"	"
D25	$2\overline{I4}$ "	31	"	"	"	"	"
D26	$2\overline{I5}$ "	32	"	"	"	"	"
D27	$2\overline{I6}$ "	33	"	"	"	"	"
D28	$2\overline{I7}$ "	34	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE: Item P07-2 shall be current limited to 5 mA.

FIGURE 8. Input breakdown current test circuit for table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input clamp diode voltage test.

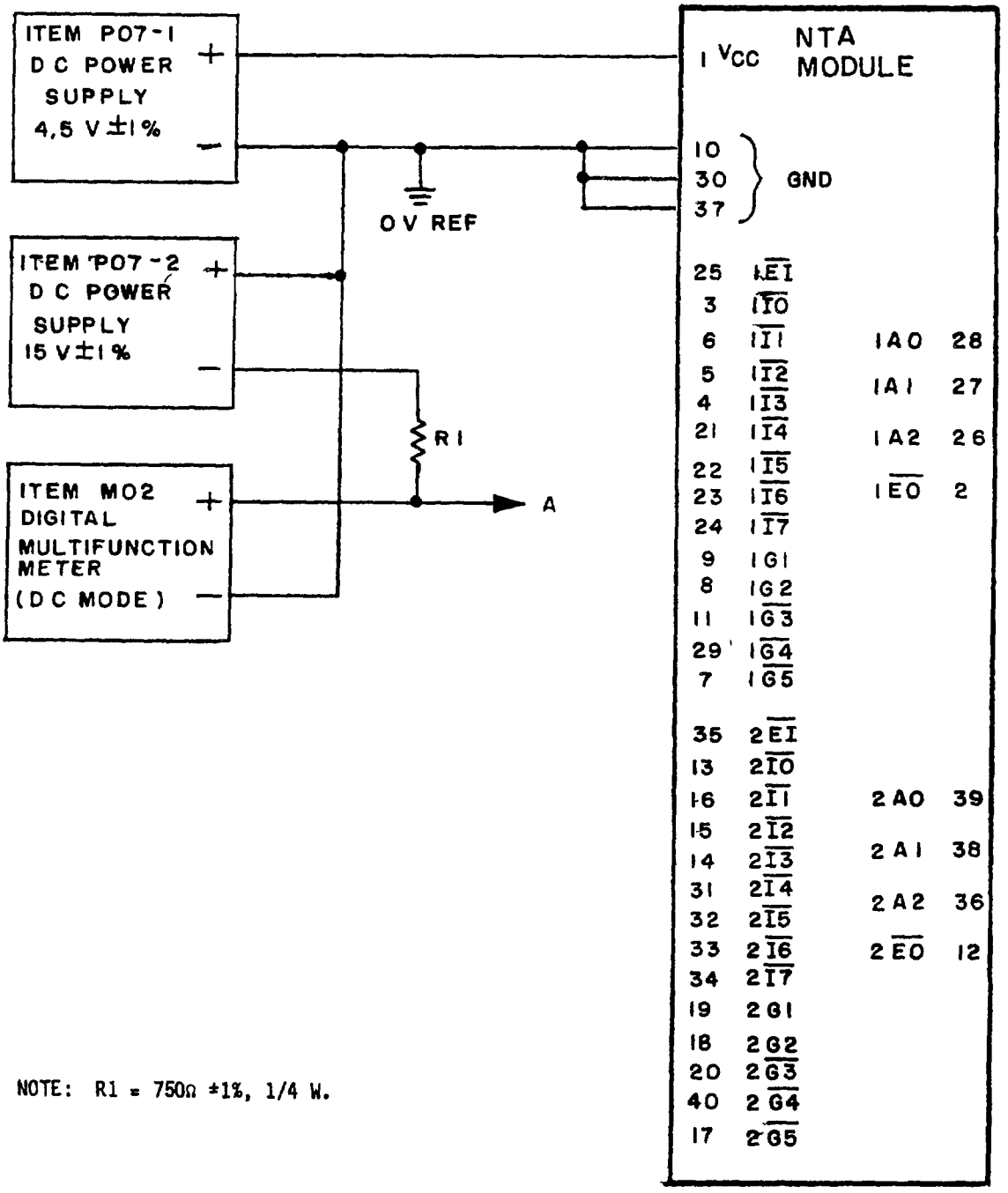
Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	$1\overline{E}T$ Input clamp diode voltage	25	-0.4	-1.5	V	9
E2	$1\overline{I}0$ "	3	"	"	"	"
E3	$1\overline{I}I$ "	6	"	"	"	"
E4	$1\overline{I}2$ "	5	"	"	"	"
E5	$1\overline{I}3$ "	4	"	"	"	"
E6	$1\overline{I}4$ "	21	"	"	"	"
E7	$1\overline{I}5$ "	22	"	"	"	"
E8	$1\overline{I}6$ "	23	"	"	"	"
E9	$1\overline{I}7$ "	24	"	"	"	"
E10	$1\overline{G}1$ "	9	"	"	"	"
E11	$1\overline{G}2$ "	8	"	"	"	"
E12	$1\overline{G}3$ "	11	"	"	"	"
E13	$1\overline{G}4$ "	29	"	"	"	"
E14	$1\overline{G}5$ "	7	"	"	"	"
E15	$2\overline{E}T$ "	35	"	"	"	"
E16	$2\overline{I}0$ "	13	"	"	"	"
E17	$2\overline{I}I$ "	16	"	"	"	"
E18	$2\overline{I}2$ "	15	"	"	"	"
E19	$2\overline{I}3$ "	14	"	"	"	"
E20	$2\overline{I}4$ "	31	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E21	$2\overline{I5}$ Input clamp diode voltage	32	-0.4	-1.5	V	9
E22	$2\overline{I6}$ "	33	"	"	"	"
E23	$2\overline{I7}$ "	34	"	"	"	"
E24	2G1 "	19	"	"	"	"
E25	2G2 "	18	"	"	"	"
E26	$2\overline{G3}$ "	20	"	"	"	"
E27	$2\overline{G4}$ "	40	"	"	"	"
E28	$2\overline{G5}$ "	17	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 750Ω ±1%, 1/4 W.

FIGURE 9. Input clamp diode voltage test circuit for table VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE IX. Output short circuit current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
F1	1A0 Output short circuit current	28	-15	-85	mA	10
F2	1A1 "	27	"	"	"	"
F3	1A2 "	26	"	"	"	"
F4	$1\overline{E0}$ "	2	"	"	"	"
F5	2A0 "	39	"	"	"	"
F6	2A1 "	38	"	"	"	"
F7	2A2 "	36	"	"	"	"
F8	$2\overline{E0}$ "	12	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

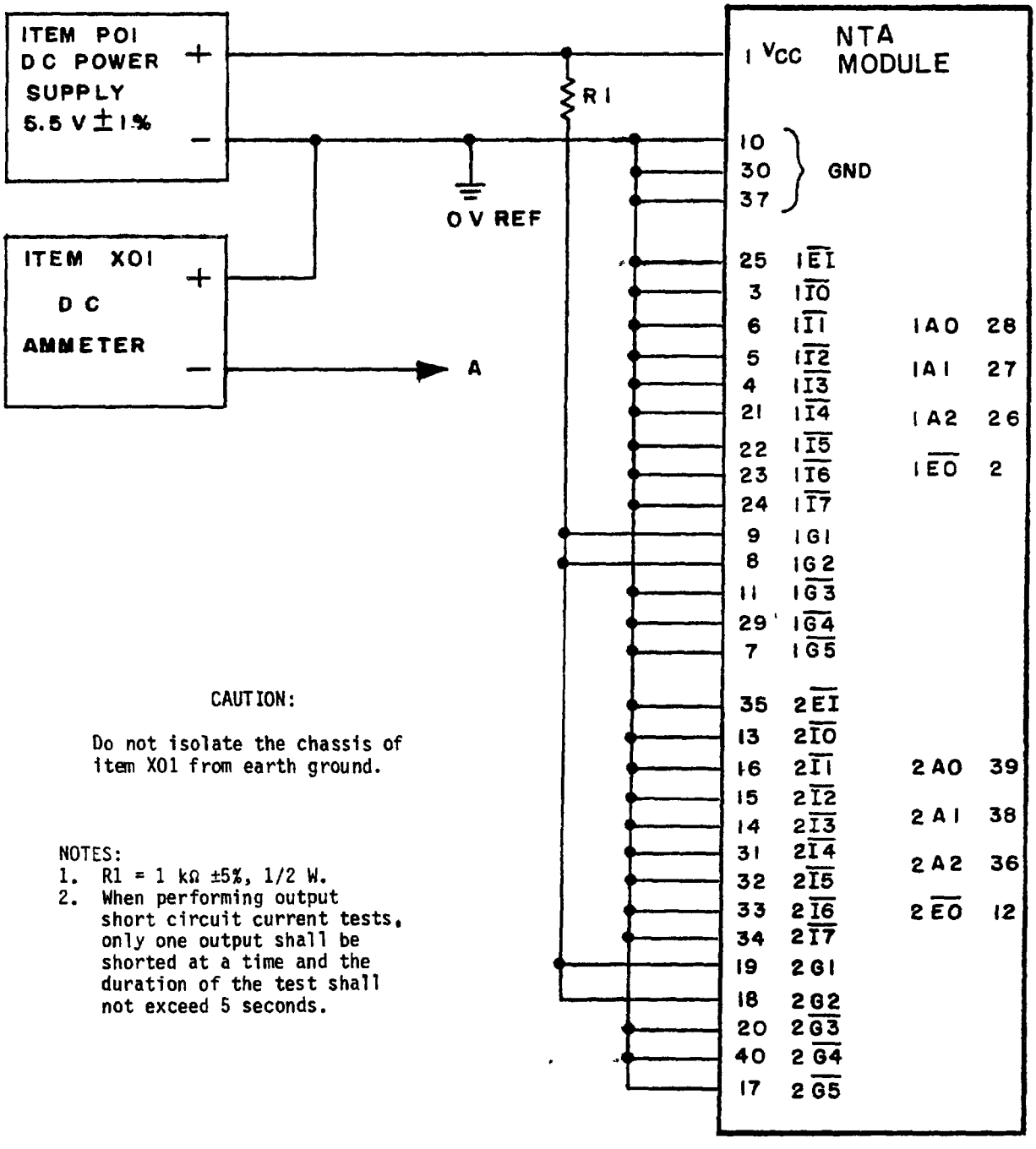


FIGURE 10. Output short circuit current test circuit for table IX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE X. Functional test.

Test condition	Parameter	Connect										Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin								Min	Max		
				28	27	26	2	39	38	36	12				
G1	Functional	25,3,6,5,4,21,22,23, 24,35,13,16,15,14,31, 32,33,34	Open	L	L	L	H	L	L	L	H	1/	1/	---	11
G2	"	25,35	3,6,5,4,21,22,23,24,13, 16,15,14,31,32,33,34	L	L	L	H	L	L	L	H	"	"	"	"
G3	"	3,6,5,4,21,22,23,24, 13,16,15,14,31,32,33, 34	25,35	L	L	L	L	L	L	L	L	"	"	"	"
G4	"	3,6,5,4,21,22,23,13, 16,15,14,31,32,33	25,35,24,34	H	H	H	H	H	H	H	H	"	"	"	"
G5	"	Open	25,3,6,5,4,21,22,23,24, 35,13,16,15,14,31,32, 33,34	H	H	H	H	H	H	H	H	"	"	"	"
G6	"	3,6,5,4,21,22,24,13, 16,15,14,31,32,34	25,35,23,33	L	H	H	H	L	H	H	H	"	"	"	"
G7	"	24,34	25,3,6,5,4,21,22,23,35, 13,16,15,14,31,32,33	L	H	H	H	L	H	H	H	"	"	"	"
G8	"	3,6,5,4,21,23,24,13, 16,15,14,31,33,34	25,35,22,32	H	L	H	H	H	L	H	H	"	"	"	"
G9	"	23,24,33,34	25,3,6,5,4,21,22,35,13, 16,15,14,31,32	H	L	H	H	H	L	H	H	"	"	"	"
G10	"	3,6,5,4,22,23,24,13, 16,15,14,32,33,34	25,35,21,31	L	L	H	H	L	L	H	H	"	"	"	"
G11	"	22,23,24,32,33,34	25,3,6,5,4,21,35,13, 16,15,14,31	L	L	H	H	L	L	H	H	"	"	"	"

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE X. Functional test - Continued.

Test condition	Parameter	Connect										Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin											
				28	27	26	2	39	38	36	12	Min	Max		
G12	Functional	3,6,5,21,22,23,24, 13,16,15,31,32,33,34	25,35,4,14	H	H	L	H	H	H	L	H	1/	1/	---	11
G13	"	21,22,23,24,31,32, 33,34	25,3,6,5,4,35,13,16, 15,14	H	H	L	H	H	H	L	H	"	"	"	"
G14	"	3,6,4,21,22,23,24, 13,16,14,31,32,33,34	25,35,5,15	L	H	L	H	L	H	L	H	"	"	"	"
G15	"	4,21,22,23,24,14,31, 32,33,34	25,3,6,5,35,13,16,15	L	H	L	H	L	H	L	H	"	"	"	"
G16	"	3,5,4,21,22,23,24,13, 15,14,31,32,33,34	25,35,6,16	H	L	L	H	H	L	L	H	"	"	"	"
G17	"	5,4,21,22,23,24,15, 14,31,32,33,34	25,3,6,35,13,16	H	L	L	H	H	L	L	H	"	"	"	"
G18	"	6,5,4,21,22,23,24,16, 15,14,31,32,33,34	25,3,35,13	L	L	L	H	L	L	L	H	"	"	"	"

1/ Output logic states:

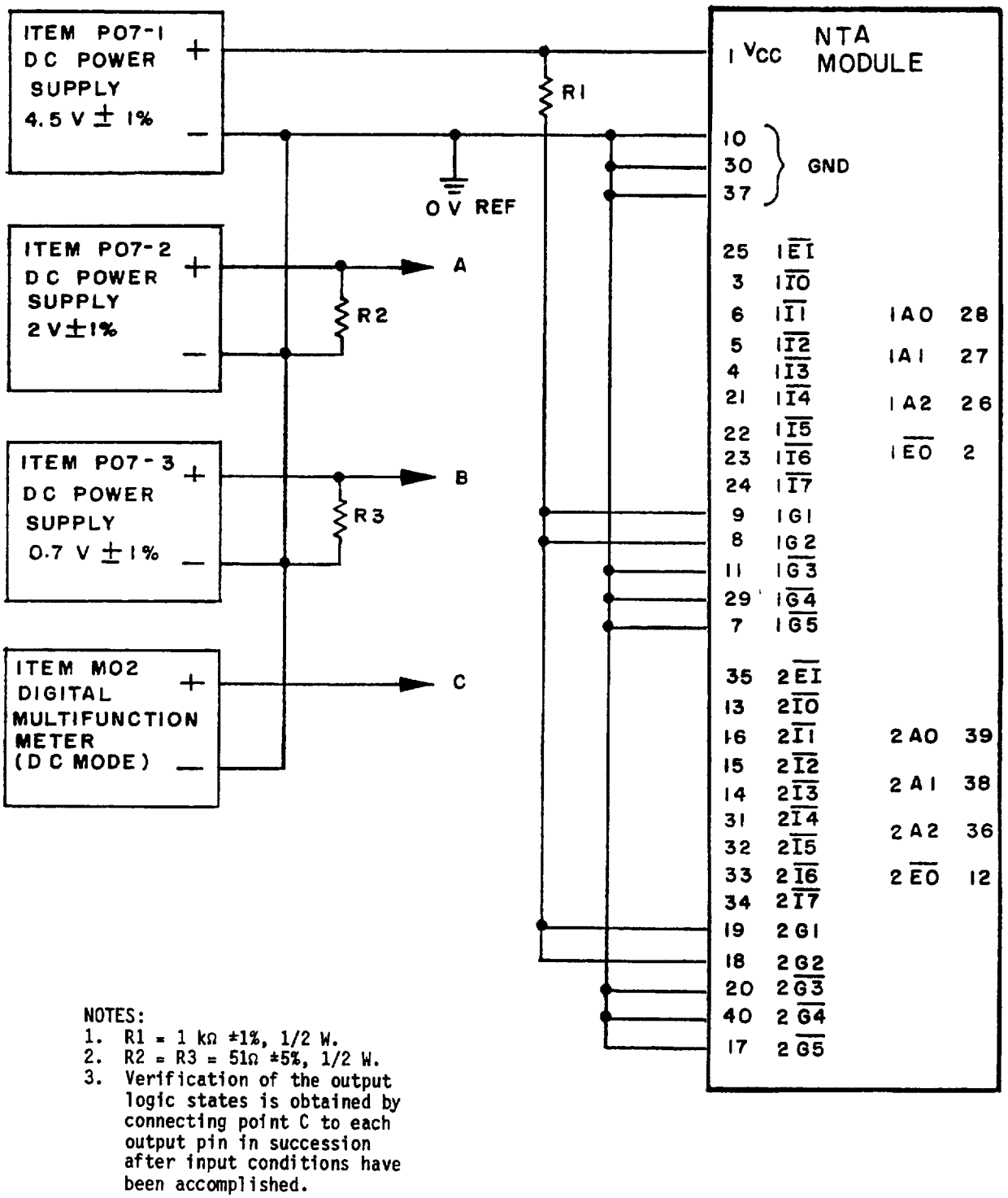
A0 - A2

Low level = 0.5 V max.
High level = 2.4 V min.

$\overline{E0}$

Low level = 0.4 V max.
High level = 2.5 V min.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 1 kΩ ±1%, 1/2 W.
2. R2 = R3 = 51Ω ±5%, 1/2 W.
3. Verification of the output logic states is obtained by connecting point C to each output pin in succession after input conditions have been accomplished.

FIGURE 11. Functional test circuit for table X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XI. Input current (high-level) test.

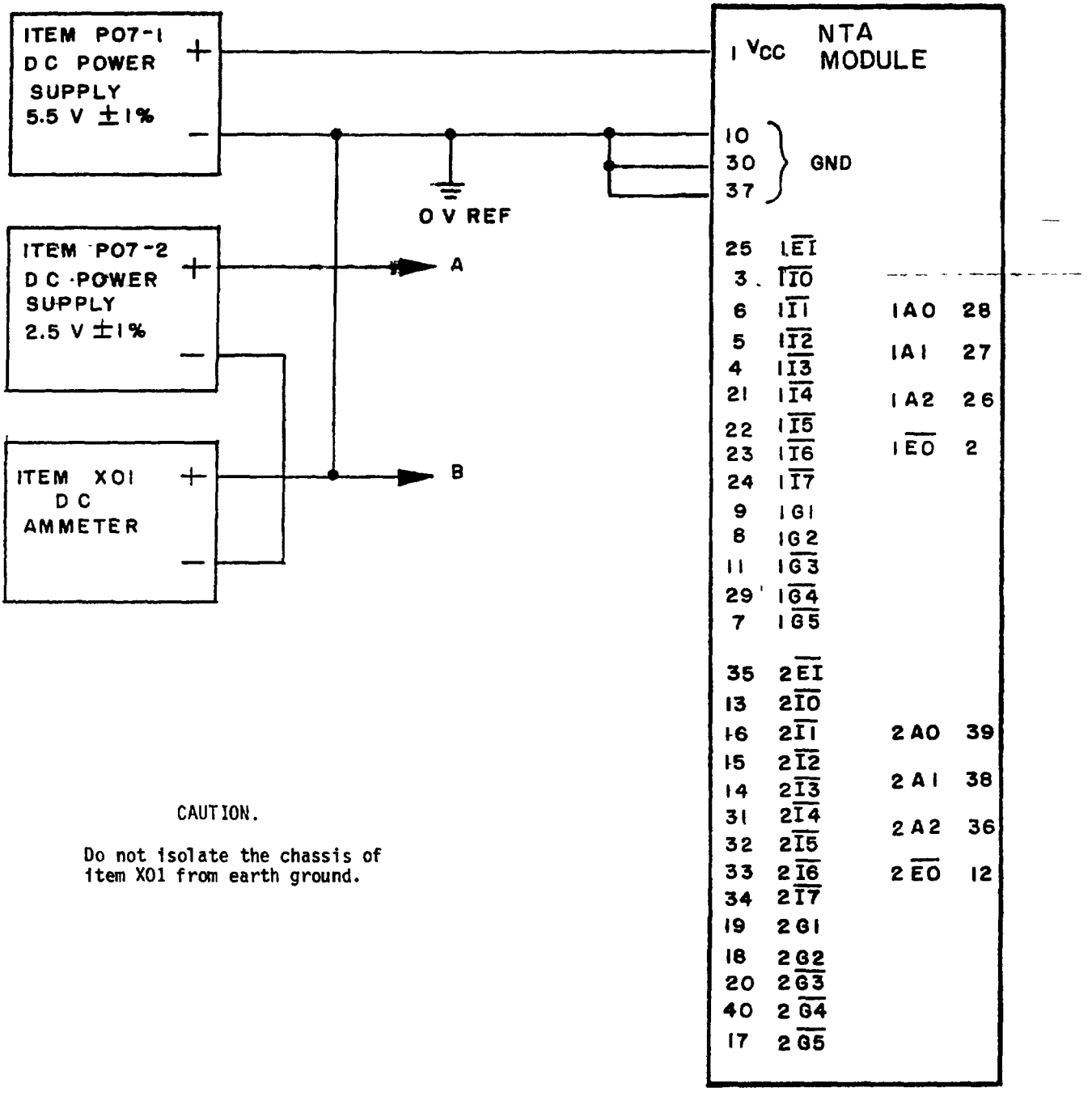
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	$1\overline{E1}$ Input current, high level	25	To all input pins except pin under test	0	20	µA	12
H2	1G1 "	9	"	"	"	"	"
H3	1G2 "	8	"	"	"	"	"
H4	$1\overline{G3}$ "	11	"	"	"	"	"
H5	$1\overline{G4}$ "	29	"	"	"	"	"
H6	$1\overline{G5}$ "	7	"	"	"	"	"
H7	$1\overline{I0}$ "	3	"	"	"	"	"
H8	$2\overline{E1}$ "	35	"	"	"	"	"
H9	2G1 "	19	"	"	"	"	"
H10	2G2 "	18	"	"	"	"	"
H11	$2\overline{G3}$ "	20	"	"	"	"	"
H12	$2\overline{G4}$ "	40	"	"	"	"	"
H13	$2\overline{G5}$ "	17	"	"	"	"	"
H14	$2\overline{I0}$ "	13	"	"	"	"	"
H15	$1\overline{I1}$ "	6	"	"	40	"	"
H16	$1\overline{I2}$ "	5	"	"	"	"	"
H17	$1\overline{I3}$ "	4	"	"	"	"	"
H18	$1\overline{I4}$ "	21	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H19	$1\overline{I15}$ Input current, high level	22	To all input pins except pin under test	0	40	µA	12
H20	$1\overline{I16}$ "	23	"	"	"	"	"
H21	$1\overline{I17}$ "	24	"	"	"	"	"
H22	$2\overline{I11}$ "	16	"	"	"	"	"
H23	$2\overline{I12}$ "	15	"	"	"	"	"
H24	$2\overline{I13}$ "	14	"	"	"	"	"
H25	$2\overline{I14}$ "	31	"	"	"	"	"
H26	$2\overline{I15}$ "	32	"	"	"	"	"
H27	$2\overline{I16}$ "	33	"	"	"	"	"
H28	$2\overline{I17}$ "	34	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



CAUTION.

Do not isolate the chassis of item XO1 from earth ground.

FIGURE 12. Input current (high-level) test circuit for table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (low-level) test.

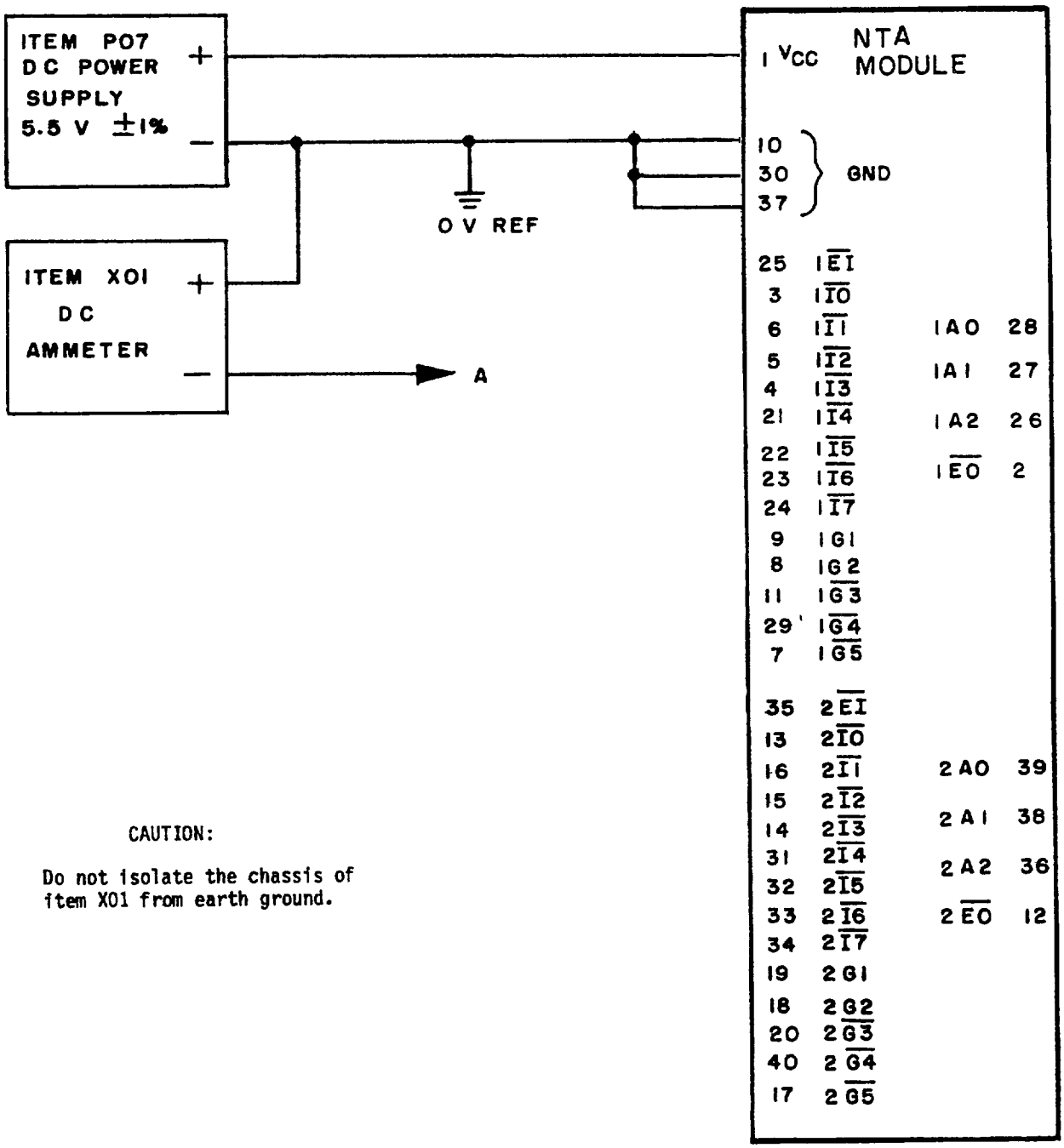
Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
J1	1ET Input current, low level	25	-10	-440	µA	13
J2	1G1 "	9	"	"	"	"
J3	1G2 "	8	"	"	"	"
J4	1G3 "	11	"	"	"	"
J5	1G4 "	29	"	"	"	"
J6	1G5 "	7	"	"	"	"
J7	1I0 "	3	"	"	"	"
J8	2ET "	35	"	"	"	"
J9	2G1 "	19	"	"	"	"
J10	2G2 "	18	"	"	"	"
J11	2G3 "	20	"	"	"	"
J12	2G4 "	40	"	"	"	"
J13	2G5 "	17	"	"	"	"
J14	2I0 "	13	"	"	"	"
J15	1IT "	6	"	-870	"	"
J16	1I2 "	5	"	"	"	"
J17	1I3 "	4	"	"	"	"
J18	1I4 "	21	"	"	"	"
J19	1I5 "	22	"	"	"	"
J20	1I6 "	23	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
J21	$1\overline{I7}$ Input current, low level	24	-10	-870	µA	13
J22	$2\overline{I1}$ "	16	"	"	"	"
J23	$2\overline{I2}$ "	15	"	"	"	"
J24	$2\overline{I3}$ "	14	"	"	"	"
J25	$2\overline{I4}$ "	31	"	"	"	"
J26	$2\overline{I5}$ "	32	"	"	"	"
J27	$2\overline{I6}$ "	33	"	"	"	"
J28	$2\overline{I7}$ "	34	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of item X01 from earth ground.

FIGURE 13. Input current (low-level) test circuit for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XIII. Output voltage (high-level) test.

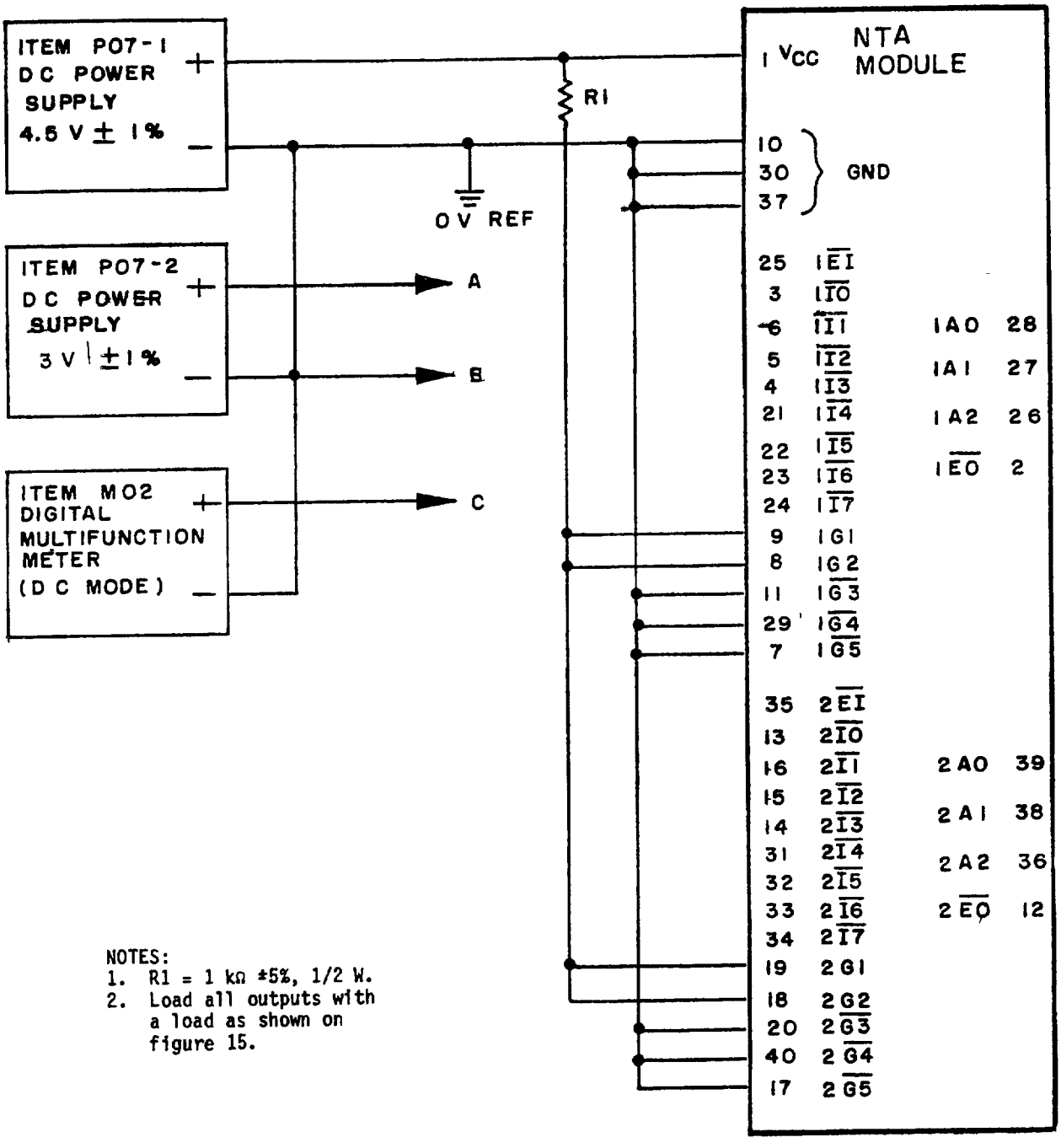
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	$1\overline{E0}$ Output voltage, high level	Open	25,24,35,34	2	2.5	4.5	V	14
K2	$2\overline{E0}$ "	"	"	12	2.5	"	"	"
K3	1A0 "	"	"	28	2.4	"	"	"
K4	1A1 "	"	"	27	"	"	"	"
K5	1A2 "	"	"	26	"	"	"	"
K6	2A0 "	"	"	39	"	"	"	"
K7	2A1 "	"	"	38	"	"	"	"
K8	2A2 "	"	"	36	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XIV. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	1EO Output voltage, low level	3,6,5,4,21,22,23, 24,13,16,15,14, 31,32,33,34	25,35	2	0	400	mV	14
L2	2EO "	"	"	12	"	400	"	"
L3	1A0 "	"	"	28	"	500	"	"
L4	1A1 "	"	"	27	"	"	"	"
L5	1A2 "	"	"	26	"	"	"	"
L6	2A0 "	"	"	39	"	"	"	"
L7	2A1 "	"	"	38	"	"	"	"
L8	2A2 "	"	"	36	"	"	"	"

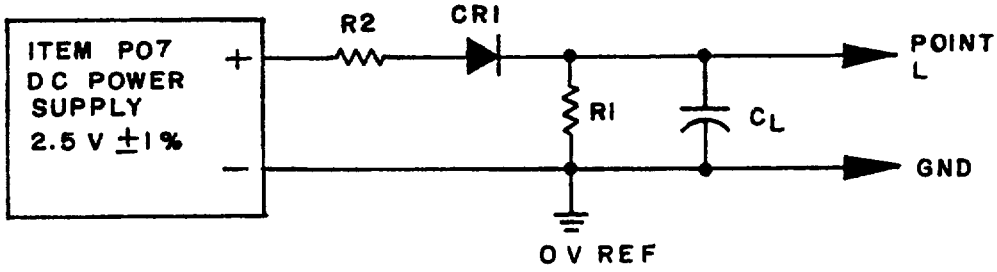
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 1 kΩ ±5%, 1/2 W.
2. Load all outputs with a load as shown on figure 15.

FIGURE 14. Output voltage (high-level and low-level) test circuit for tables XIII and XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. CR1 = 1N3600, 1N4150 or equivalent.
2. R1 = 2.4 kΩ ±1%, 1/8 W for outputs A0-A2.
   = 5.69 kΩ ±1%, 1/8 W for outputs $\overline{E0}$.
3. R2 = Select a value for a current flow of 12 mA ±5% out-of-point L with point L held at 0.5 V for outputs A0-A2.
   = Select a value for a current flow of 4 mA ±5% out-of-point L with point L held at 0.4 V for outputs $\overline{E0}$.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions Q, R, U, V, W, and X.
   = 150 ±5 pF (including probe and parasitic capacitance) for test conditions S and T.
5. C_L is not required for test conditions K and L.
6. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

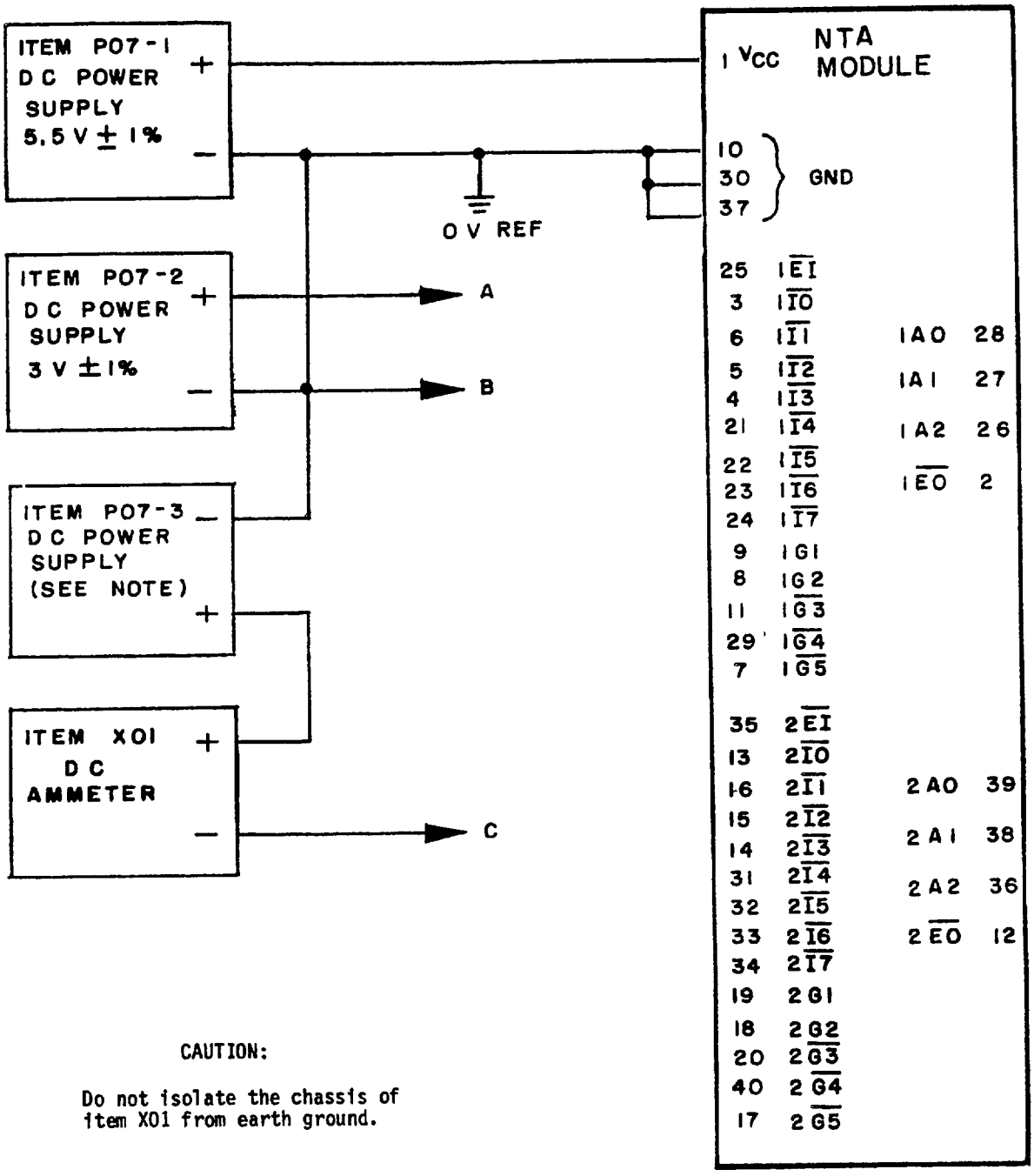
FIGURE 15. NTA logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XV. High impedance leakage current (high-level and low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	High impedance leakage current, high level	8,11,29,7,18,20, 40,17	9,19	28	-1	20	µA	16
M2	"	"	"	27	"	"	"	"
M3	"	"	"	26	"	"	"	"
M4	"	"	"	39	"	"	"	"
M5	"	"	"	38	"	"	"	"
M6	"	"	"	36	"	"	"	"
N1	High impedance leakage current, low level	"	"	28	-20	1	"	"
N2	"	"	"	27	"	"	"	"
N3	"	"	"	26	"	"	"	"
N4	"	"	"	39	"	"	"	"
N5	"	"	"	38	"	"	"	"
N6	"	"	"	36	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTE:
Adjust item P07-3 voltages as follows:
2.4 V ±1% for test conditions M1-M6.
0.5 V ±1% for test conditions N1-N6.

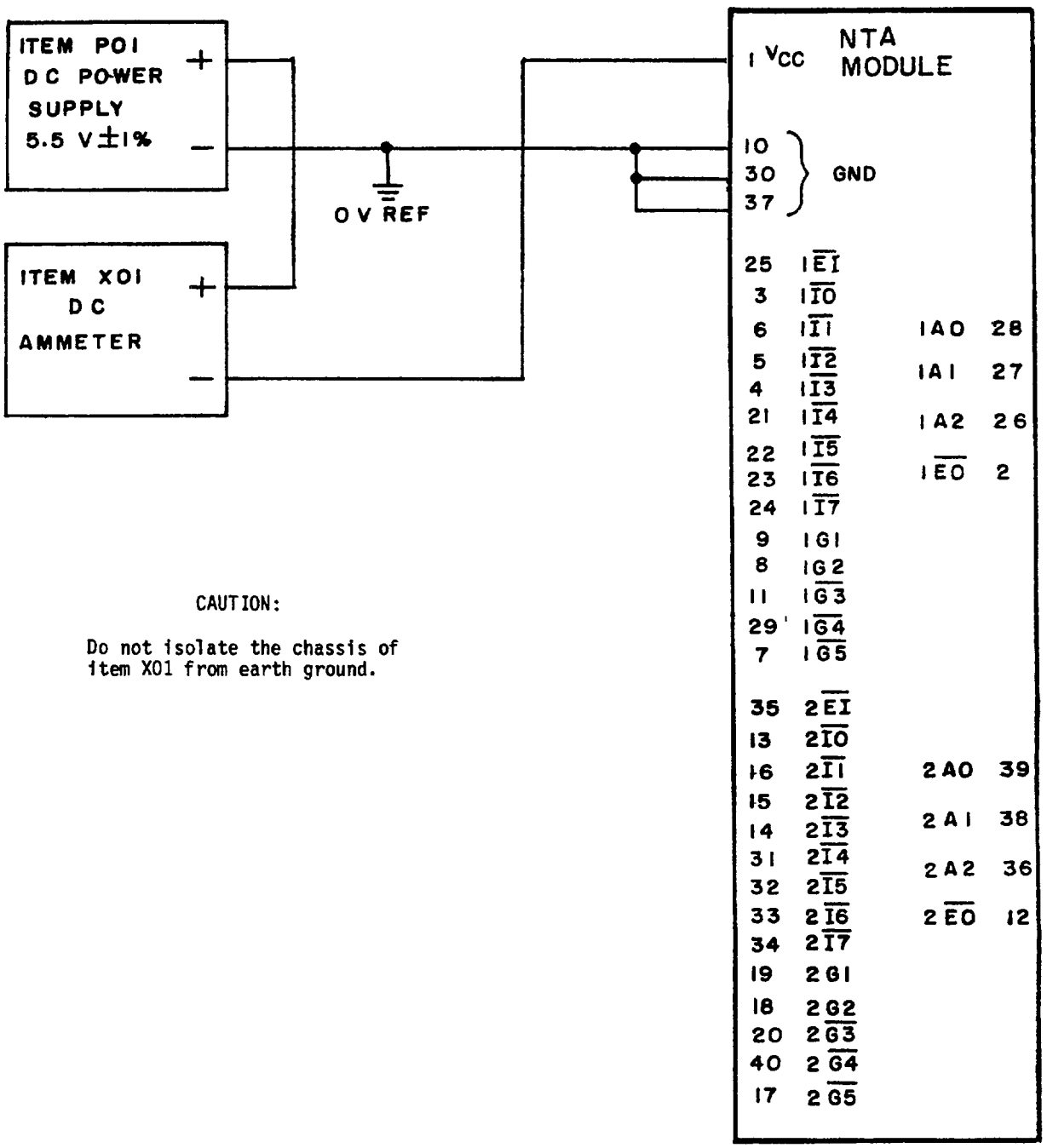
FIGURE 16. High impedance leakage current (high-level and low-level) test circuit for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVI. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
P	Power supply current	5	48	mA	17

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

FIGURE 17. Power supply current test circuit for table XVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	Propagation delay time (t_{PLH}), out-of-phase, C_L = 50 pF	3,6,5,4,21,22,23	25,35	24	2	3	15	ns	18,19
Q2	"	3,6,5,4,21,22,24	"	23	"	"	"	"	"
Q3	"	3,6,5,4,21,23,24	"	22	"	"	"	"	"
Q4	"	3,6,5,4,22,23,24	"	21	"	"	"	"	"
Q5	"	3,6,5,21,22,23,24	"	4	"	"	"	"	"
Q6	"	3,6,4,21,22,23,24	"	5	"	"	"	"	"
Q7	"	3,5,4,21,22,23,24	"	6	"	"	"	"	"
Q8	"	6,5,4,21,22,23,24	"	3	"	"	"	"	"
Q9	"	13,16,15,14,31,32,33	"	34	12	"	"	"	"
Q10	"	13,16,15,14,31,32,34	"	33	"	"	"	"	"
Q11	"	13,16,15,14,31,33,34	"	32	"	"	"	"	"
Q12	"	13,16,15,14,32,33,34	"	31	"	"	"	"	"
Q13	"	13,16,15,31,32,33,34	"	14	"	"	"	"	"
Q14	"	13,16,14,31,32,33,34	"	15	"	"	"	"	"
Q15	"	13,15,14,31,32,33,34	"	16	"	"	"	"	"
Q16	"	16,15,14,31,32,33,34	"	13	"	"	"	"	"
Q17	Propagation delay time (t_{PLH}), inphase, C_L = 50 pF	3,6,5,4,21,22,23,24	Open	25	2	5	21	"	"
Q18	Propagation delay time (t_{PLH}), inphase, C_L = 50 pF	13,16,15,14,31,32,33,34	Open	35	12	5	21	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q19	Propagation delay time (t_{PLH}), out-of-phase, C_L = 50 pF	3,6,5,4,21,22,23	24,34	25	28	6	22	ns	18,19
Q20	"	"	"	"	27	"	"	"	"
Q21	"	"	"	"	26	"	"	"	"
Q22	"	13,16,15,14,31,32,33	"	35	39	"	"	"	"
Q23	"	"	"	"	38	"	"	"	"
Q24	"	"	"	"	36	"	"	"	"
Q25	"	3,6,5,4,21,22,23	25,35	24	28	5	21	"	"
Q26	"	"	"	"	27	"	"	"	"
Q27	"	"	"	"	26	"	"	"	"
Q28	"	13,16,15,14,31,32,33	"	34	39	"	"	"	"
Q29	"	"	"	"	38	"	"	"	"
Q30	"	"	"	"	36	"	"	"	"
Q31	"	3,6,5,4,21,22,24	"	23	27	"	"	"	"
Q32	"	3,6,5,4,21,22,24	"	23	26	"	"	"	"
Q33	"	13,16,15,14,31,32,34	"	33	38	"	"	"	"
Q34	"	13,16,15,14,31,32,34	"	33	36	"	"	"	"
Q35	"	3,6,5,4,21,23,24	"	22	28	"	"	"	"
Q36	"	3,6,5,4,21,23,24	"	22	26	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q37	Propagation delay time (t_{PLH}), out-of-phase, C_L = 50 pF	13,16,15,14,31,33,34	25,35	32	39	5	21	ns	18,19
Q38	"	13,16,15,14,31,33,34	"	32	36	"	"	"	"
Q39	"	3,6,5,4,22,23,24	"	21	26	"	"	"	"
Q40	"	13,16,15,14,32,33,34	"	31	36	"	"	"	"
Q41	"	3,6,5,21,22,23,24	"	4	28	"	"	"	"
Q42	"	3,6,5,21,22,23,24	"	4	27	"	"	"	"
Q43	"	13,16,15,31,32,33,34	"	14	39	"	"	"	"
Q44	"	13,16,15,31,32,33,34	"	14	38	"	"	"	"
Q45	"	3,6,4,21,22,23,24	"	5	27	"	"	"	"
Q46	"	13,16,14,31,32,33,34	"	15	38	"	"	"	"
Q47	"	3,5,4,21,22,23,24	"	6	28	"	"	"	"
Q48	"	13,15,14,31,32,33,34	"	16	39	"	"	"	"
Q49	Propagation delay time (t_{PLH}), inphase, C_L = 50 pF	3,6,5,4,21,24	25,35,22	23	28	8	29	"	"
Q50	"	13,16,15,14,31,34	25,35,32	33	39	"	"	"	"
Q51	"	3,6,5,21,23,24	25,35,4,14	22	27	"	"	"	"
Q52	"	13,16,15,31,33,34	"	32	38	"	"	"	"
Q53	"	3,6,5,22,23,24	"	21	28	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q54	Propagation delay time (t_{PLH}), inphase, C_L = 50 pF	3,6,5,22,23,24	25,35,4,14	21	27	8	29	ns	18,19
Q55	"	13,16,15,32,33,34	"	31	39	"	"	"	"
Q56	"	13,16,15,32,33,34	"	31	38	"	"	"	"
Q57	"	3,4,21,22,23,24	25,35,6	5	28	"	"	"	"
Q58	"	13,14,31,32,33,34	25,35,16	15	39	"	"	"	"
R1	Propagation delay time (t_{PHL}), out-of-phase, C_L = 50 pF	3,6,5,4,21,22,23	25,35	24	2	12	40	"	"
R2	"	3,6,5,4,21,22,24	"	23	"	"	"	"	"
R3	"	3,6,5,4,21,23,24	"	22	"	"	"	"	"
R4	"	3,6,5,4,22,23,24	"	21	"	"	"	"	"
R5	"	3,6,5,21,22,23,24	"	4	"	"	"	"	"
R6	"	3,6,4,21,22,23,24	"	5	"	"	"	"	"
R7	"	3,5,4,21,22,23,24	"	6	"	"	"	"	"
R8	"	6,5,4,21,22,23,24	"	3	"	"	"	"	"
R9	"	13,16,15,14,31,32,33	"	34	12	"	"	"	"
R10	"	13,16,15,14,31,32,34	"	33	"	"	"	"	"
R11	"	13,16,15,14,31,33,34	"	32	"	"	"	"	"
R12	"	13,16,15,14,32,33,34	"	31	"	"	"	"	"
R13	"	13,16,15,31,32,33,34	"	14	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R14	Propagation delay time (t_{PHL}), out-of-phase, C_L = 50 pF	13,16,14,31,32,33,34	25,35	15	12	12	40	ns	18,19
R15	"	13,15,14,31,32,33,34	"	16	"	"	"	"	"
R16	"	16,15,14,31,32,33,34	"	13	"	"	"	"	"
R17	Propagation delay time (t_{PHL}), inphase, C_L = 50 pF	3,6,5,4,21,22,23,24	Open	25	2	11	38	"	"
R18	Propagation delay time (t_{PHL}), inphase, C_L = 50 pF	13,16,15,14,31,32,33,34	Open	35	12	11	38	"	"
R19	Propagation delay time (t_{PHL}), out-of-phase, C_L = 50 pF	3,6,5,4,21,22,23	24,34	25	28	7	25	"	"
R20	"	"	"	"	27	"	"	"	"
R21	"	"	"	"	26	"	"	"	"
R22	"	13,16,15,14,31,32,33	"	35	39	"	"	"	"
R23	"	"	"	"	38	"	"	"	"
R24	"	"	"	"	36	"	"	"	"
R25	"	3,6,5,4,21,22,23	25,35	24	28	6	22	"	"
R26	"	"	"	"	27	"	"	"	"
R27	"	"	"	"	26	"	"	"	"
R28	"	13,16,15,14,31,32,33	"	34	39	"	"	"	"
R29	"	"	"	"	38	"	"	"	"
R30	"	"	"	"	36	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R31	Propagation delay time (t_{PHL}), out-of-phase, C_L = 50 pF	3,6,5,4,21,22,24	25,35	23	27	6	22	ns	18,19
R32	"	3,6,5,4,21,22,24	"	23	26	"	"	"	"
R33	"	13,16,15,14,31,32,34	"	33	38	"	"	"	"
R34	"	13,16,15,14,31,32,34	"	33	36	"	"	"	"
R35	"	3,6,5,4,21,23,24	"	22	28	"	"	"	"
R36	"	3,6,5,4,21,23,24	"	22	26	"	"	"	"
R37	"	13,16,15,14,31,33,34	"	32	39	"	"	"	"
R38	"	13,16,15,14,31,33,34	"	32	36	"	"	"	"
R39	"	3,6,5,4,22,23,24	"	21	26	"	"	"	"
R40	"	13,16,15,14,32,33,34	"	31	36	"	"	"	"
R41	"	3,6,5,21,22,23,24	"	4	28	"	"	"	"
R42	"	3,6,5,21,22,23,24	"	4	27	"	"	"	"
R43	"	13,16,15,31,32,33,34	"	14	39	"	"	"	"
R44	"	13,16,15,31,32,33,34	"	14	38	"	"	"	"
R45	"	3,6,4,21,22,23,24	"	5	27	"	"	"	"
R46	"	13,16,14,31,32,33,34	"	15	38	"	"	"	"
R47	"	3,5,4,21,22,23,24	"	6	28	"	"	"	"
R48	"	13,15,14,31,32,33,34	"	16	39	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R49	Propagation delay time (t_{PHL}), inphase, C_L = 50 pF	3,6,5,4,21,24	25,35,22	23	28	8	29	ns	18,19
R50	"	13,16,15,14,31,34	25,35,32	33	39	"	"	"	"
R51	"	3,6,5,21,23,24	25,35,4,14	22	27	"	"	"	"
R52	"	13,16,15,31,33,34	"	32	38	"	"	"	"
R53	"	3,6,5,22,23,24	"	21	28	"	"	"	"
R54	"	3,6,5,22,23,24	"	21	27	"	"	"	"
R55	"	13,16,15,32,33,34	"	31	39	"	"	"	"
R56	"	13,16,15,32,33,34	"	31	38	"	"	"	"
R57	"	3,4,21,22,23,24	25,35,6	5	28	"	"	"	"
R58	"	13,14,31,32,33,34	25,35,16	15	39	"	"	"	"
S1	Propagation delay time (t_{PLH}), out-of-phase, C_L = 150 pF	3,6,5,4,21,22,23	25,35	24	2	3	25	"	"
S2	"	3,6,5,4,21,22,24	"	23	"	"	"	"	"
S3	"	3,6,5,4,21,23,24	"	22	"	"	"	"	"
S4	"	3,6,5,4,22,23,24	"	21	"	"	"	"	"
S5	"	3,6,5,21,22,23,24	"	4	"	"	"	"	"
S6	"	3,6,4,21,22,23,24	"	5	"	"	"	"	"
S7	"	3,5,4,21,22,23,24	"	6	"	"	"	"	"
S8	"	6,5,4,21,22,23,24	"	3	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S9	Propagation delay time (t_{PLH}), out-of-phase, C_L = 150 pF	13,16,15,14,31,32,33	25,35	34	12	3	25	ns	18,19
S10	"	13,16,15,14,31,32,34	"	33	"	"	"	"	"
S11	"	13,16,15,14,31,33,34	"	32	"	"	"	"	"
S12	"	13,16,15,14,32,33,34	"	31	"	"	"	"	"
S13	"	13,16,15,31,32,33,34	"	14	"	"	"	"	"
S14	"	13,16,14,31,32,33,34	"	15	"	"	"	"	"
S15	"	13,15,14,31,32,33,34	"	16	"	"	"	"	"
S16	"	16,15,14,31,32,33,34	"	13	"	"	"	"	"
S17	Propagation delay time (t_{PLH}), inphase, C_L = 150 pF	3,6,5,4,21,22,23,24	Open	25	2	5	31	"	"
S18	Propagation delay time (t_{PLH}), inphase, C_L = 150 pF	13,16,15,14,31,32,33,34	Open	35	12	5	31	"	"
S19	Propagation delay time (t_{PLH}), out-of-phase, C_L = 150 pF	3,6,5,4,21,22,23	24,34	25	28	6	32	"	"
S20	"	"	"	"	27	"	"	"	"
S21	"	"	"	"	26	"	"	"	"
S22	"	13,16,15,14,31,32,33	"	35	39	"	"	"	"
S23	"	"	"	"	38	"	"	"	"
S24	"	"	"	"	36	"	"	"	"
S25	"	3,6,5,4,21,22,23	25,35	24	28	6	31	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S26	Propagation delay time (t_{PLH}), out-of-phase, C_L = 150 pF	3,6,5,4,21,22,23	25,35	24	27	6	31	ns	18,19
S27	"	3,6,5,4,21,22,23	"	24	26	"	"	"	"
S28	"	13,16,15,14,31,32,33	"	34	39	"	"	"	"
S29	"	"	"	"	38	"	"	"	"
S30	"	"	"	"	36	"	"	"	"
S31	"	3,6,5,4,21,22,24	"	23	27	"	"	"	"
S32	"	3,6,5,4,21,22,24	"	23	26	"	"	"	"
S33	"	13,16,15,14,31,32,34	"	33	38	"	"	"	"
S34	"	13,16,15,14,31,32,34	"	33	36	"	"	"	"
S35	"	3,6,5,4,21,23,24	"	22	28	"	"	"	"
S36	"	3,6,5,4,21,23,24	"	22	26	"	"	"	"
S37	"	13,16,15,14,31,33,34	"	32	39	"	"	"	"
S38	"	13,16,15,14,31,33,34	"	32	36	"	"	"	"
S39	"	3,6,5,4,22,23,24	"	21	26	"	"	"	"
S40	"	13,16,15,14,32,33,34	"	31	36	"	"	"	"
S41	"	3,6,5,21,22,23,24	"	4	28	"	"	"	"
S42	"	3,6,5,21,22,23,24	"	4	27	"	"	"	"
S43	"	13,16,15,31,32,33,34	"	14	39	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S44	Propagation delay time (t_{PLH}), out-of-phase, C_L = 150 pF	13,16,15,31,32,33,34	25,35	14	38	6	31	ns	18,19
S45	"	3,6,4,21,22,23,24	"	5	27	"	"	"	"
S46	"	13,16,14,31,32,33,34	"	15	38	"	"	"	"
S47	"	3,5,4,21,22,23,24	"	6	28	"	"	"	"
S48	"	13,15,14,31,32,33,34	"	16	39	"	"	"	"
S49	Propagation delay time (t_{PLH}), inphase, C_L = 150 pF	3,6,5,4,21,24	25,35,22	23	28	8	39	"	"
S50	"	13,16,15,14,31,34	25,35,32	33	39	"	"	"	"
S51	"	3,6,5,21,23,24	25,35,4,14	22	27	"	"	"	"
S52	"	13,16,15,31,33,34	"	32	38	"	"	"	"
S53	"	3,6,5,22,23,24	"	21	28	"	"	"	"
S54	"	3,6,5,22,23,24	"	21	27	"	"	"	"
S55	"	13,16,15,32,33,34	"	31	39	"	"	"	"
S56	"	13,16,15,32,33,34	"	31	38	"	"	"	"
S57	"	3,4,21,22,23,24	25,35,6	5	28	"	"	"	"
S58	"	13,14,31,32,33,34	25,35,16	15	39	"	"	"	"
T1	Propagation delay time (t_{PHL}), out-of-phase, C_L = 150 pF	3,6,5,4,21,22,23	25,35	24	2	12	50	"	"
T2	Propagation delay time (t_{PHL}) out-of-phase, C_L = 150 pF	3,6,5,4,21,22,24	25,35	23	2	12	50	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T3	Propagation delay time (t_{PHL}), out-of-phase, C_L = 150 pF	3,6,5,4,21,23,24	25,35	22	2	12	50	ns	18,19
T4	"	3,6,5,4,22,23,24	"	21	"	"	"	"	"
T5	"	3,6,5,21,22,23,24	"	4	"	"	"	"	"
T6	"	3,6,4,21,22,23,24	"	5	"	"	"	"	"
T7	"	3,5,4,21,22,23,24	"	6	"	"	"	"	"
T8	"	6,5,4,21,22,23,24	"	3	"	"	"	"	"
T9	"	13,16,15,14,31,32,33	"	34	12	"	"	"	"
T10	"	13,16,15,14,31,32,34	"	33	"	"	"	"	"
T11	"	13,16,15,14,31,33,34	"	32	"	"	"	"	"
T12	"	13,16,15,14,32,33,34	"	31	"	"	"	"	"
T13	"	13,16,15,31,32,33,34	"	14	"	"	"	"	"
T14	"	13,16,14,31,32,33,34	"	15	"	"	"	"	"
T15	"	13,15,14,31,32,33,34	"	16	"	"	"	"	"
T16	"	16,15,14,31,32,33,34	"	13	"	"	"	"	"
T17	Propagation delay time (t_{PHL}), inphase, C_L = 150 pF	3,6,5,4,21,22,23,24	Open	25	2	11	48	"	"
T18	Propagation delay time (t_{PHL}), inphase, C_L = 150 pF	13,16,15,14,31,32,33,34	Open	35	12	11	48	"	"
T19	Propagation delay time (t_{PHL}), out-of-phase, C_L = 150 pF	3,6,5,4,21,22,23	24,34	25	28	7	35	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T20	Propagation delay time (t_{PHL}), out-of-phase, C_L = 150 pF	3,6,5,4,21,22,23	24,34	25	27	7	35	ns	18,19
T21	"	3,6,5,4,21,22,23	"	25	26	"	"	"	"
T22	"	13,16,15,14,31,32,33	"	35	39	"	"	"	"
T23	"	"	"	"	38	"	"	"	"
T24	"	"	"	"	36	"	"	"	"
T25	"	3,6,5,4,21,22,23	25,35	24	28	6	32	"	"
T26	"	"	"	"	27	"	"	"	"
T27	"	"	"	"	26	"	"	"	"
T28	"	13,16,15,14,31,32,33	"	34	39	"	"	"	"
T29	"	"	"	"	38	"	"	"	"
T30	"	"	"	"	36	"	"	"	"
T31	"	3,6,5,4,21,22,24	"	23	27	"	"	"	"
T32	"	3,6,5,4,21,22,24	"	23	26	"	"	"	"
T33	"	13,16,15,14,31,32,34	"	33	38	"	"	"	"
T34	"	13,16,15,14,31,32,34	"	33	36	"	"	"	"
T35	"	3,6,5,4,21,23,24	"	22	28	"	"	"	"
T36	"	3,6,5,4,21,23,24	"	22	26	"	"	"	"
T37	"	13,16,15,14,31,33,34	"	32	39	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T38	Propagation delay time (t_{PHL}), out-of-phase, C_L = 150 pF	13,16,15,14,31,33,34	25,35	32	36	6	32	ns	18,19
T39	"	3,6,5,4,22,23,24	"	21	26	"	"	"	"
T40	"	13,16,15,14,32,33,34	"	31	36	"	"	"	"
T41	"	3,6,5,21,22,23,24	"	4	28	"	"	"	"
T42	"	3,6,5,21,22,23,24	"	4	27	"	"	"	"
T43	"	13,16,15,31,32,33,34	"	14	39	"	"	"	"
T44	"	13,16,15,31,32,33,34	"	14	38	"	"	"	"
T45	"	3,6,4,21,22,23,24	"	5	27	"	"	"	"
T46	"	13,16,14,31,32,33,34	"	15	38	"	"	"	"
T47	"	3,5,4,21,22,23,24	"	6	28	"	"	"	"
T48	"	13,15,14,31,32,33,34	"	16	39	"	"	"	"
T49	Propagation delay time (t_{PHL}), inphase, C_L = 150 pF	3,6,5,4,21,24	25,35,22	23	28	8	39	"	"
T50	"	13,16,15,14,31,34	25,35,32	33	39	"	"	"	"
T51	"	3,6,5,21,23,24	25,35,4,14	22	27	"	"	"	"
T52	"	13,16,15,31,33,34	"	32	38	"	"	"	"
T53	"	3,6,5,22,23,24	"	21	28	"	"	"	"
T54	"	3,6,5,22,23,24	"	21	27	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T55	Propagation delay time (t_{PHL}), inphase, C_L = 150 pF	13,16,15,32,33,34	25,35,4,14	31	39	8	39	ns	18,19
T56	"	13,16,15,32,33,34	25,35,4,14	31	38	"	"	"	"
T57	"	3,4,21,22,23,24	25,35,6	5	28	"	"	"	"
T58	"	13,14,31,32,33,34	25,35,16	15	39	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

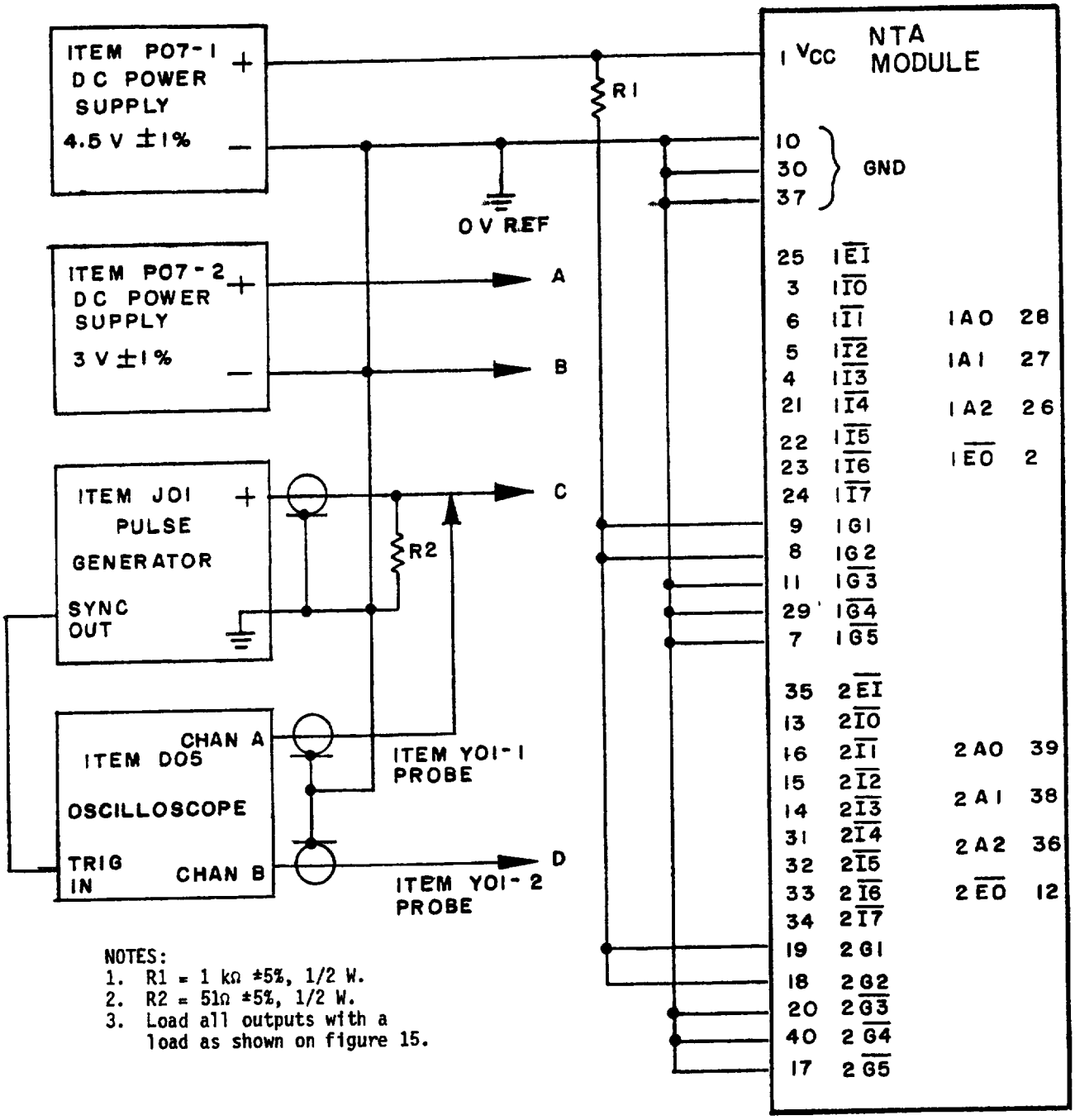
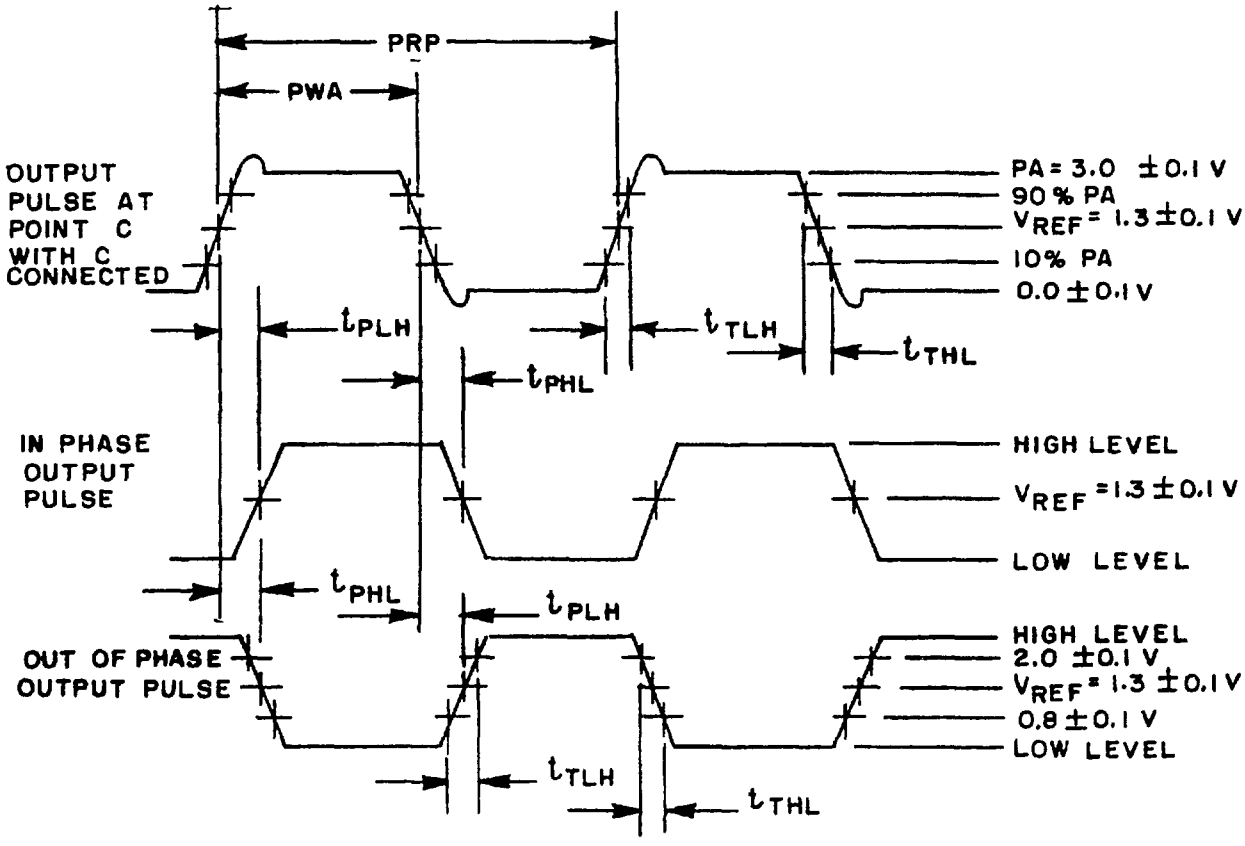


FIGURE 18. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PWA = 200 ±20 ns.
   PRP = 400 ±40 ns.
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns.
2. Before taking the specified measurement, verify that the output pulse high level is $\geq$ 2.5 V for the $\overline{E0}$ outputs and $\geq$ 2.4 V for the A0-A2 outputs and the low level is $\leq$ 0.4 V for the $\overline{E0}$ outputs and $\leq$ 0.5 V for the A0-A2 outputs.

FIGURE 19. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Disable recovery time (t_{PHZ} and t_{PLZ}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	Disable recovery time, t_{PHZ}	8,3,6,5,4,21,22,23	24,11,29,7	9	28	8	60	ns	20,21
U2	"	"	"	"	27	"	"	"	"
U3	"	"	"	"	26	"	"	"	"
U4	"	9,3,6,5,4,21,22,23	"	8	28	"	"	"	"
U5	"	"	"	"	27	"	"	"	"
U6	"	"	"	"	26	"	"	"	"
U7	"	18,13,16,15,14,31,32,33	34,20,40,17	19	39	"	"	"	"
U8	"	"	"	"	38	"	"	"	"
U9	"	"	"	"	36	"	"	"	"
U10	"	19,13,16,15,14,31,32,33	"	18	39	"	"	"	"
U11	"	"	"	"	38	"	"	"	"
U12	"	"	"	"	36	"	"	"	"
U13	"	9,8,3,6,5,4,21,22,23	24,29,7	11	28	"	55	"	20,22
U14	"	"	"	"	27	"	"	"	"
U15	"	"	"	"	26	"	"	"	"
U16	"	"	24,11,7	29	28	"	"	"	"
U17	"	"	"	"	27	"	"	"	"
U18	"	"	"	"	26	"	"	"	"
U19	"	"	24,11,29	7	28	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Disable recovery time (t_{PHZ} and t_{PLZ}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U20	Disable recovery time, t_{PHZ}	9,8,3,6,5,4,21,22,23	24,11,29	7	27	8	55	ns	20,22
U21	"	9,8,3,6,5,4,21,22,23	24,11,29	7	26	"	"	"	"
U22	"	19,18,13,16,15,14,31,32,33	34,40,17	20	39	"	"	"	"
U23	"	"	"	"	38	"	"	"	"
U24	"	"	"	"	36	"	"	"	"
U25	"	"	34,20,17	40	39	"	"	"	"
U26	"	"	"	"	38	"	"	"	"
U27	"	"	"	"	36	"	"	"	"
U28	"	"	34,20,40	17	39	"	"	"	"
U29	"	"	"	"	38	"	"	"	"
U30	"	"	"	"	36	"	"	"	"
U31	Disable recovery time, t_{PLZ}	8,3,6,5,4,21,22,23,24	11,29,7	9	28	9	60	"	20,21
U32	"	"	"	"	27	"	"	"	"
U33	"	"	"	"	26	"	"	"	"
U34	"	9,3,6,5,4,21,22,23,24	"	8	28	"	"	"	"
U35	"	"	"	"	27	"	"	"	"
U36	"	"	"	"	26	"	"	"	"
U37	"	18,13,16,15,14,31,32,33,34	20,40,17	19	39	"	"	"	"
U38	"	18,13,16,15,14,31,32,33,34	20,40,17	19	38	"	"	"	"

Source: https://assist.dla.mil - Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Disable recovery time (t_{PHZ} and t_{PLZ}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U39	Disable recovery time, t_{PLZ}	18,13,16,15,14,31,32,33,34	20,40,17	19	36	9	60	ns	20,21
U40	"	19,13,16,15,14,31,32,33,34	"	18	39	"	"	"	"
U41	"	"	"	"	38	"	"	"	"
U42	"	"	"	"	36	"	"	"	"
U43	"	9,8,3,6,5,4,21,22,23,24	29,7	11	28	"	55	"	20,22
U44	"	"	"	"	27	"	"	"	"
U45	"	"	"	"	26	"	"	"	"
U46	"	"	11,7	29	28	"	"	"	"
U47	"	"	"	"	27	"	"	"	"
U48	"	"	"	"	26	"	"	"	"
U49	"	"	11,29	7	28	"	"	"	"
U50	"	"	"	"	27	"	"	"	"
U51	"	"	"	"	26	"	"	"	"
U52	"	19,18,13,16,15,14,31,32,33,34	40,17	20	39	"	"	"	"
U53	"	"	"	"	38	"	"	"	"
U54	"	"	"	"	36	"	"	"	"
U55	"	"	20,17	40	39	"	"	"	"
U56	"	"	"	"	38	"	"	"	"
U57	"	"	"	"	36	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Disable recovery time (t_{PHZ} and t_{PLZ}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U58	Disable recovery time, t_{PLZ}	19,18,13,16,15,14,31,32,33,34	20,40	17	39	9	55	ns	20,22
U59	"	"	"	"	38	"	"	"	"
U60	"	"	"	"	36	"	"	"	"

Source: https://assist.dla.mil - Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XIX. Enable recovery time (t_{PZH} and t_{PZL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	Enable recovery time, t_{PZH}	8,3,6,5,4,21,22,23	24,11,29,7	9	28	11	49	ns	20,21
V2	"	"	"	"	27	"	"	"	"
V3	"	"	"	"	26	"	"	"	"
V4	"	9,3,6,5,4,21,22,23	"	8	28	"	"	"	"
V5	"	"	"	"	27	"	"	"	"
V6	"	"	"	"	26	"	"	"	"
V7	"	18,13,16,15,14,31,32,33	34,20,40,17	19	39	"	"	"	"
V8	"	"	"	"	38	"	"	"	"
V9	"	"	"	"	36	"	"	"	"
V10	"	19,13,16,15,14,31,32,33	"	18	39	"	"	"	"
V11	"	"	"	"	38	"	"	"	"
V12	"	"	"	"	36	"	"	"	"
V13	"	9,8,3,6,5,4,21,22,23	24,29,7	11	28	10	43	"	20,22
V14	"	"	"	"	27	"	"	"	"
V15	"	"	"	"	26	"	"	"	"
V16	"	"	24,11,7	29	28	"	"	"	"
V17	"	"	"	"	27	"	"	"	"
V18	"	"	"	"	26	"	"	"	"
V19	"	"	24,11,29	7	28	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XIX. Enable recovery time (t_{PZH} and t_{PZL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V20	Enable recovery time, t_{PZH}	9,8,3,6,5,4,21,22,23	24,11,29	7	27	10	43	ns	20,22
V21	"	9,8,3,6,5,4,21,22,23	24,11,29	7	26	"	"	"	"
V22	"	19,18,13,16,15,14,31,32,33	34,40,17	20	39	"	"	"	"
V23	"	"	"	"	38	"	"	"	"
V24	"	"	"	"	36	"	"	"	"
V25	"	"	34,20,17	40	39	"	"	"	"
V26	"	"	"	"	38	"	"	"	"
V27	"	"	"	"	36	"	"	"	"
V28	"	"	34,20,40	17	39	"	"	"	"
V29	"	"	"	"	38	"	"	"	"
V30	"	"	"	"	36	"	"	"	"
V31	Enable recovery time, t_{PZL}	8,3,6,5,4,21,22,23,24	11,29,7	9	28	"	49	ns	20,21
V32	"	"	"	"	27	"	"	"	"
V33	"	"	"	"	26	"	"	"	"
V34	"	9,3,6,5,4,21,22,23,24	"	8	28	"	"	"	"
V35	"	"	"	"	27	"	"	"	"
V36	"	"	"	"	26	"	"	"	"
V37	"	18,13,16,15,14,31,32,33,34	20,40,17	19	39	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XIX. Enable recovery time (t_{PZH} and t_{PZL}) test - Continued.

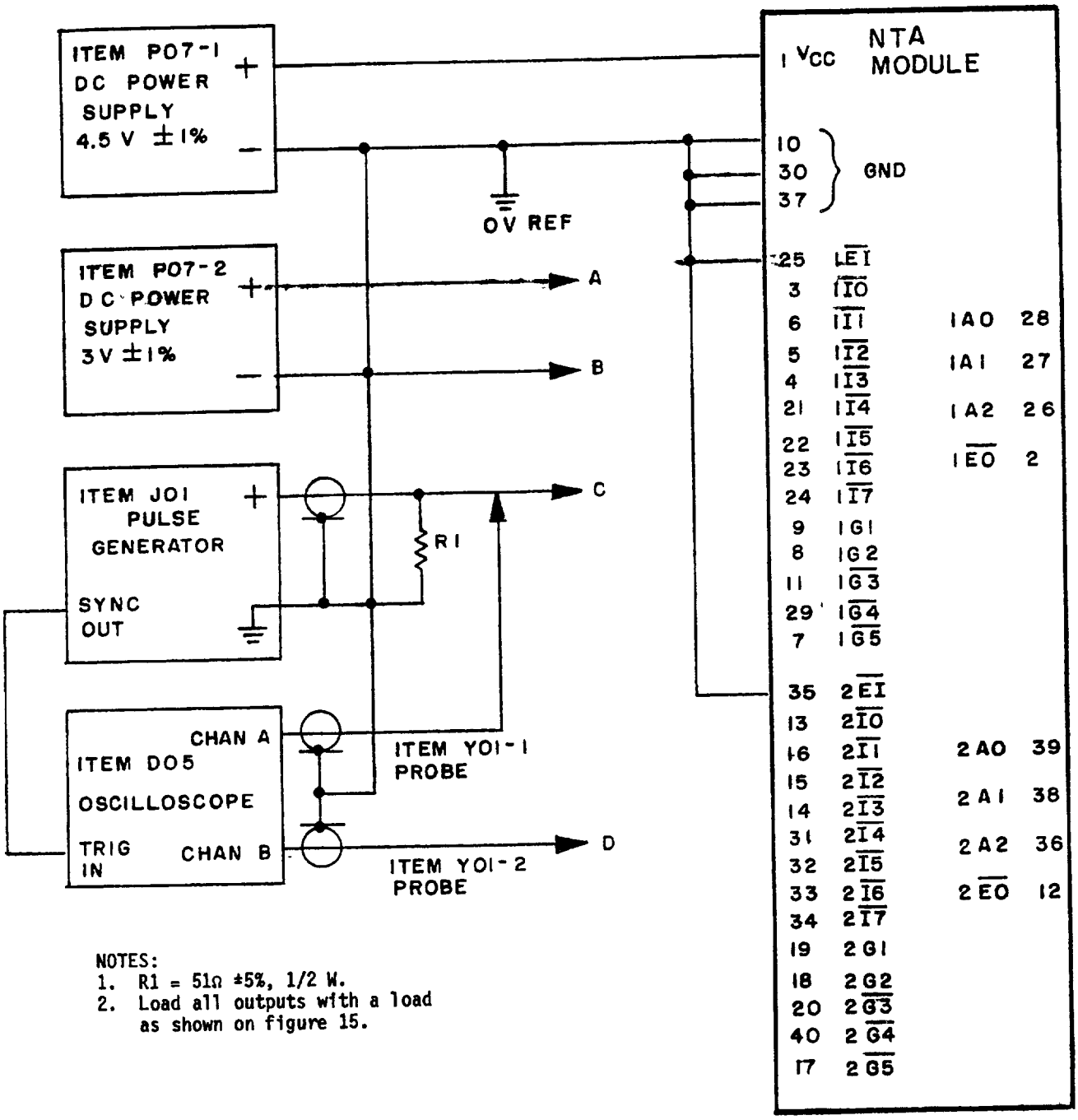
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V38	Enable recovery time, t_{PZL}	18,13,16,15,14,31,32,33,34	20,40,17	19	38	10	49	ns	20,21
V39	"	18,13,16,15,14,31,32,33,34	"	19	36	"	"	"	"
V40	"	19,13,16,15,14,31,32,33,34	"	18	39	"	"	"	"
V41	"	"	"	"	38	"	"	"	"
V42	"	"	"	"	36	"	"	"	"
V43	"	9,8,3,6,5,4,21,22,23,24	29,7	11	28	9	43	"	20,22
V44	"	"	"	"	27	"	"	"	"
V45	"	"	"	"	26	"	"	"	"
V46	"	"	11,7	29	28	"	"	"	"
V47	"	"	"	"	27	"	"	"	"
V48	"	"	"	"	26	"	"	"	"
V49	"	"	11,29	7	28	"	"	"	"
V50	"	"	"	"	27	"	"	"	"
V51	"	"	"	"	26	"	"	"	"
V52	"	19,18,13,16,15,14,31,32,33,34	40,17	20	39	"	"	"	"
V53	"	"	"	"	38	"	"	"	"
V54	"	"	"	"	36	"	"	"	"
V55	"	"	20,17	40	39	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XIX. Enable recovery time (t_{PZH} and t_{PZL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V56	Enable recovery time, t_{PZL}	19,18,13,16,15,14,31,32,33,34	20,17	40	38	9	43	ns	20,22
V57	"	"	20,17	40	36	"	"	"	"
V58	"	"	20,40	17	39	"	"	"	"
V59	"	"	"	"	38	"	"	"	"
V60	"	"	"	"	36	"	"	"	"

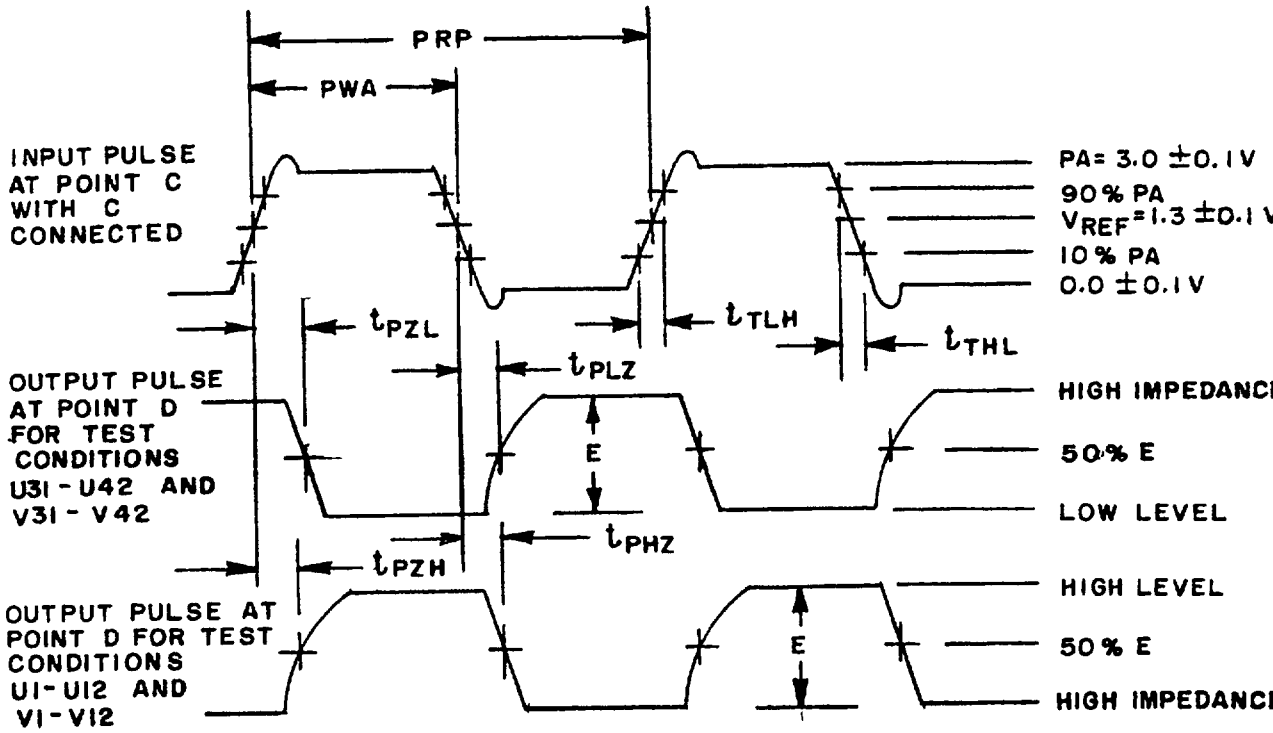
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load all outputs with a load as shown on figure 15.

FIGURE 20. Disable recovery time (t_{PHZ} and t_{PLZ}) and enable recovery time (t_{PZH} and t_{PZL}) test circuit for tables XVIII and XIX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

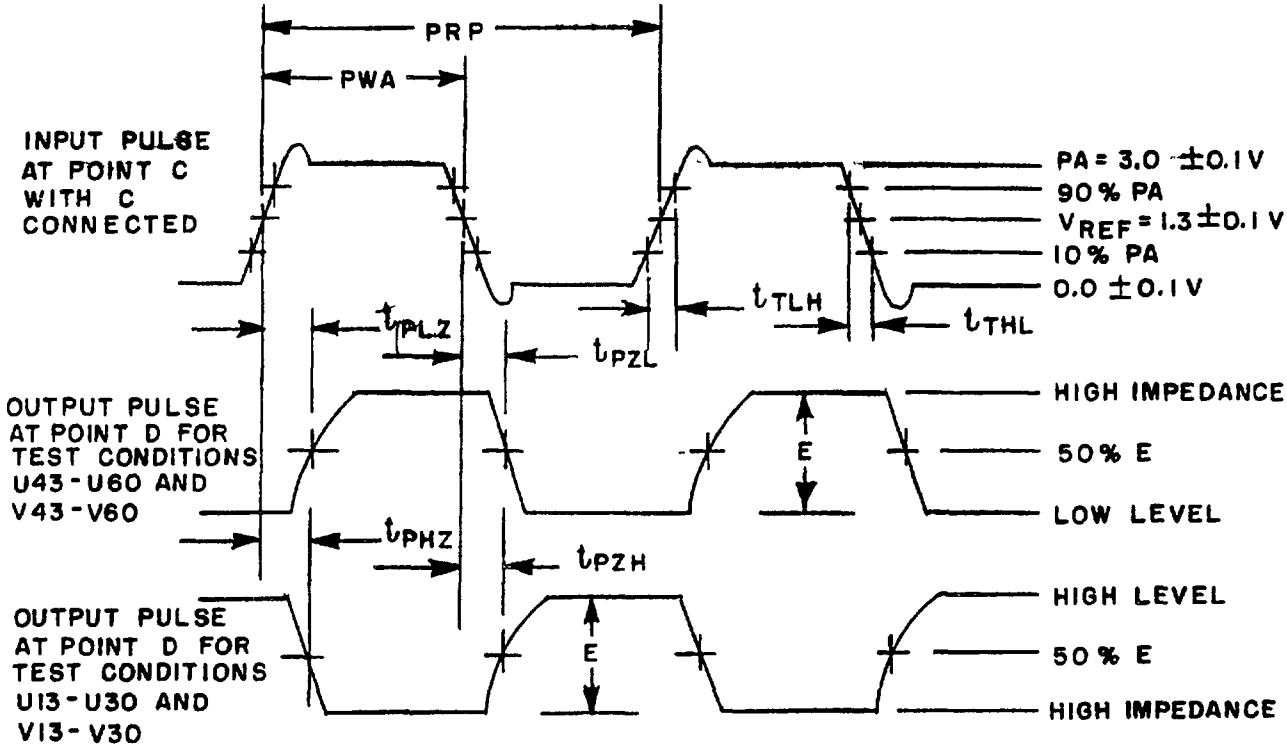


NOTES:
1. Input pulse parameters:
   PWA = 200 ±20 ns.
   PRP = 400 ±40 ns.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns.

2. Before making the specified measurements, verify that the output low level is $\leq$ 0.5 V and the output high level is $\geq$ 2.4 V.

FIGURE 21. Disable recovery time (t_{PHZ} and t_{PLZ}) and enable recovery time (t_{PZH} and t_{PZL}) test circuit waveforms for tables XVIII and XIX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PWA = 200 ±20 ns.
   PRP = 400 ±40 ns.
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns.

2. Before making the specified measurements, verify that the output low level is $\leq$ 0.5 V and the output high level is $\geq$ 2.4 V.

FIGURE 22. Disable recovery time (t_{PHZ} and t_{PLZ}) and enable recovery time (t_{PZH} and t_{PZL}) test circuit waveforms for tables XVIII and XIX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XX. Output crosstalk (low-level and high-level) test.

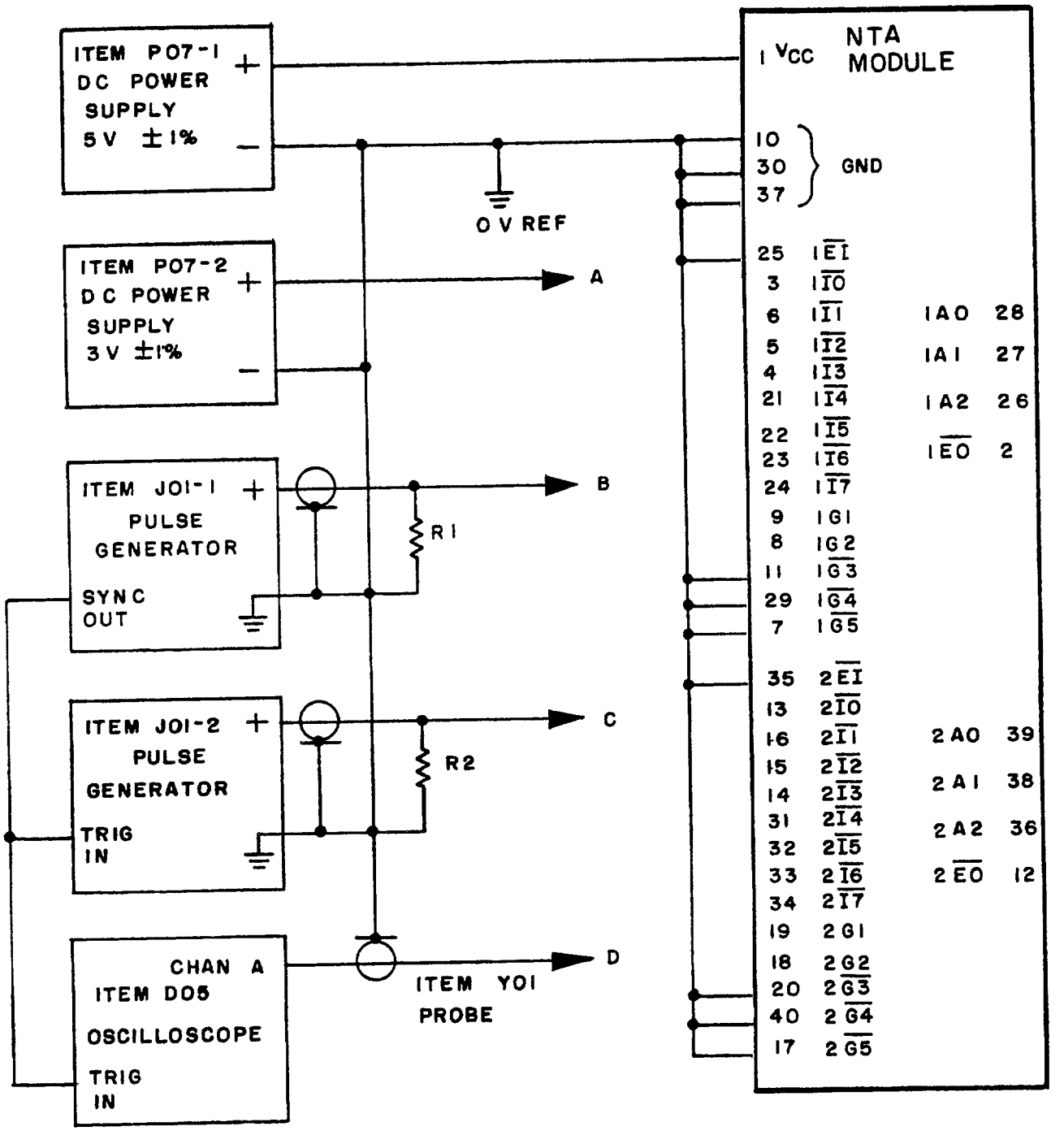
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W1	Output crosstalk, low level	9,8,19,18,6,5,4,21,22,24,16,15, 14,31,32,34	3,13	23,33	28	0	400	mV	23,24
W2	"	9,8,19,18,6,5,4,21,22,24,16,15, 14,31,32,34	"	"	39	"	"	"	"
W3	"	9,8,19,18,6,5,4,21,23,24,16,15, 14,31,33,34	"	23,32	27	"	"	"	"
W4	"	9,8,19,18,6,5,4,21,22,24,16,15, 14,31,33,34	"	23,32	38	"	"	"	"
W5	"	9,8,19,18,6,5,21,22,23,24,16, 15,31,32,33,34	"	4,14	26	"	"	"	"
W6	"	9,8,19,18,6,5,21,22,23,24,16, 15,31,32,33,34	"	4,14	36	"	"	"	"
X1	Output crosstalk, high level	9,8,19,18,6,5,4,21,22,23,16,15, 14,31,32,33	"	24,34	2	0	"	"	"
X2	"	9,8,19,18,6,5,4,21,22,23,16,15, 14,31,32,33	"	"	12	"	"	"	"
X3	"	9,8,19,18,3,5,4,21,22,23,13,15, 14,31,32,33	6,16	"	28	"	"	"	"
X4	"	9,8,19,18,3,5,4,21,22,23,13,15, 14,31,32,33	6,16	"	39	"	"	"	"
X5	"	9,8,19,18,3,6,4,21,22,23,13,16, 14,31,32,33	5,15	"	27	"	"	"	"
X6	"	9,8,19,18,3,6,4,21,22,23,24,13, 16,14,31,32,33,34	5,15	"	38	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XX. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
X7	Output crosstalk, high level	9,8,19,18,3,6,5,4,,22,23,13,16, 15,14,32,33	21,31	24,34	26	0	400	mV	23,24
X8	Output crosstalk, high level	9,8,19,18,3,6,5,22,23,13,16, 15,14,32,33	21,31	24,34	36	0	400	mV	23,24

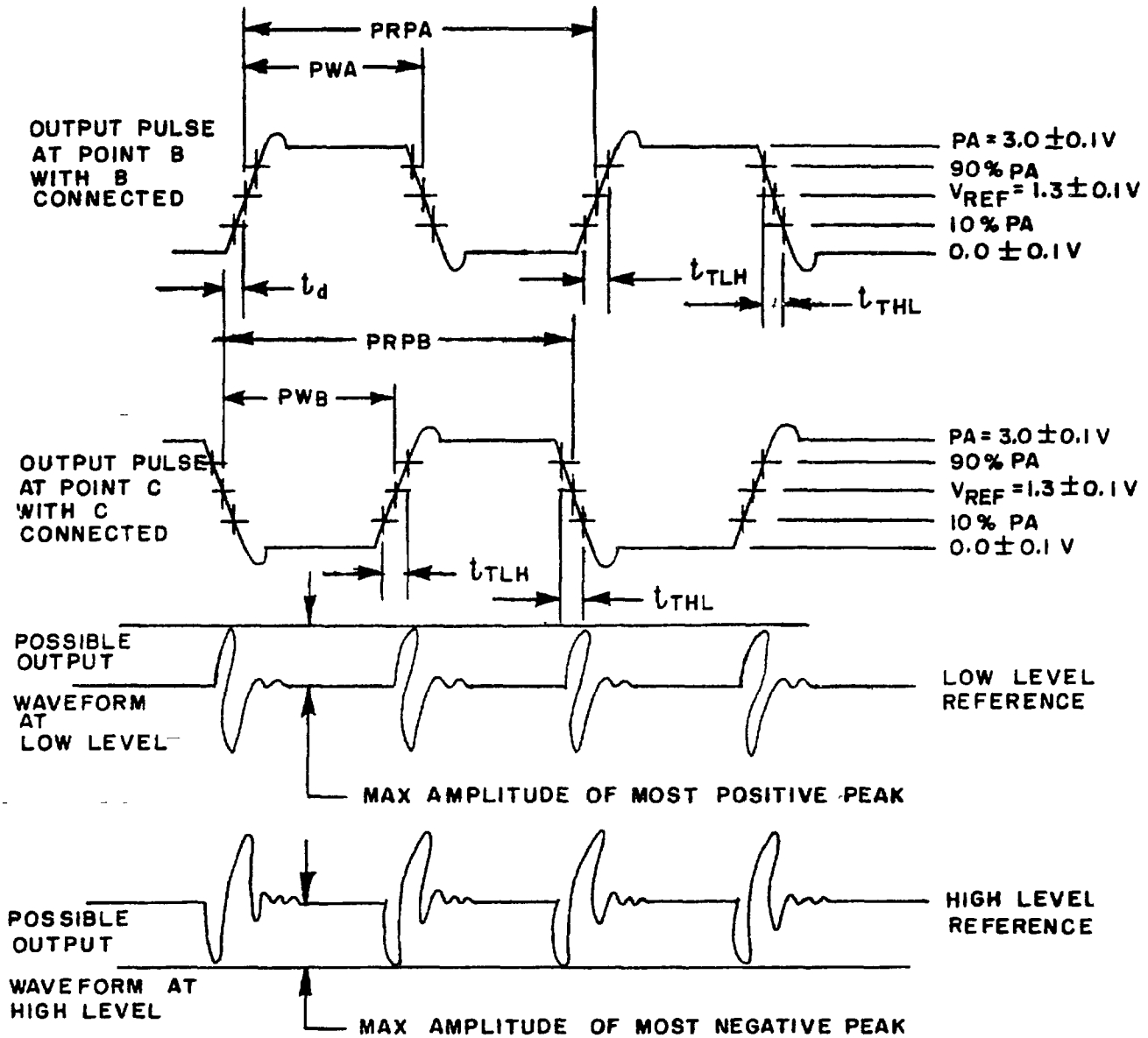
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with a load as shown on figure 15.

FIGURE 23. Output crosstalk (low-level and high-level) test circuit for table XX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PWA = PWB = 200 ±20 ns.
   PRPA = PRPB = 400 ±40 ns.
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns.
   $t_d = 0 \pm 2$ ns.
2. The lead length of electrical conductors between resistors R1 and R2 and input pins, item Y01 to input and output pins, load circuits to output pins, and ground lead of item Y01 to test fixture and module ground pins shall be less then 2 inches (50.8 mm).

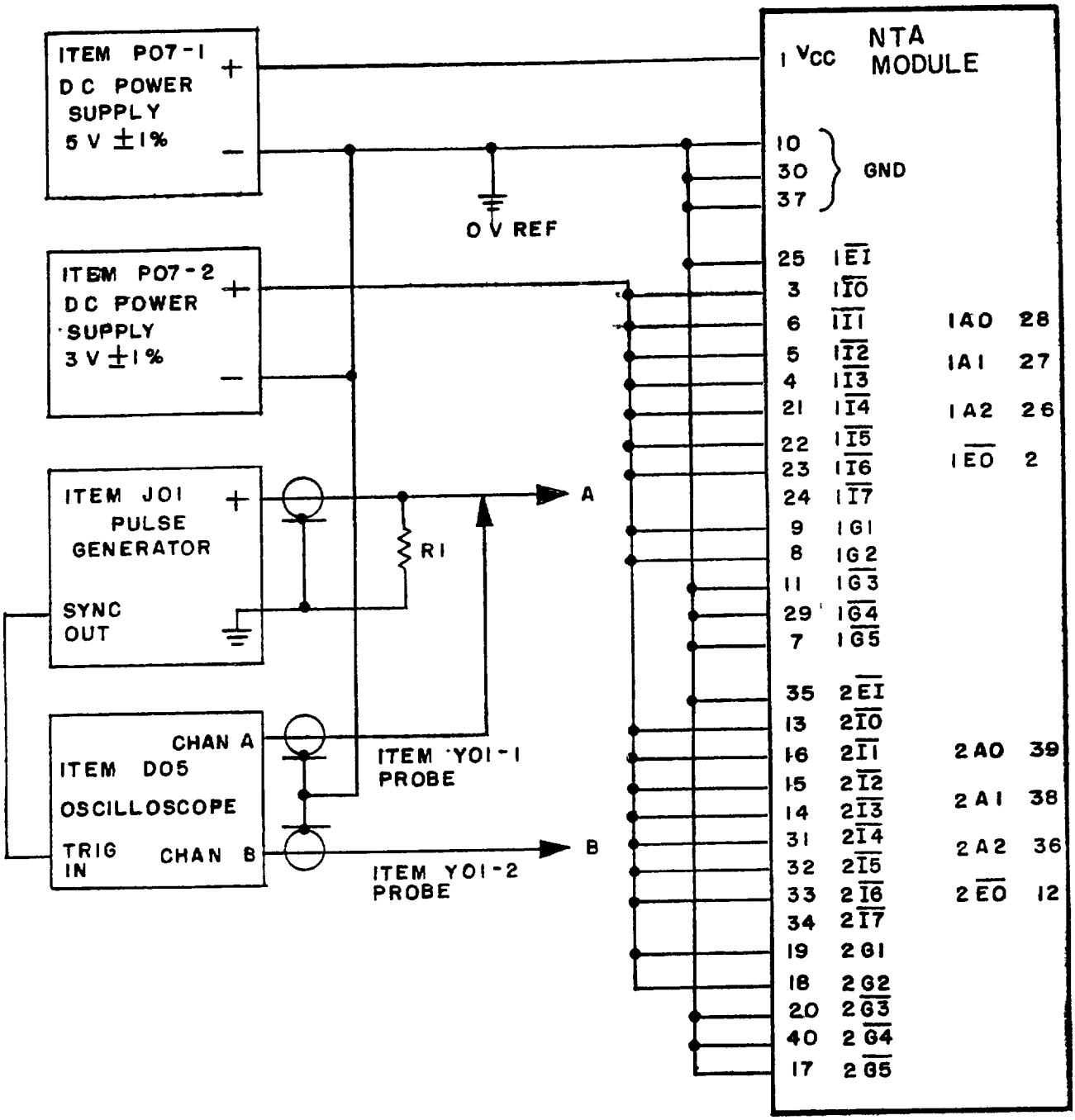
FIGURE 24. Output crosstalk (low-level and high-level) test circuit waveforms for table XX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XXI. Power supply transient amplitude test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y	Power supply transient amplitude	24,34	1	0	400	mV	25,26

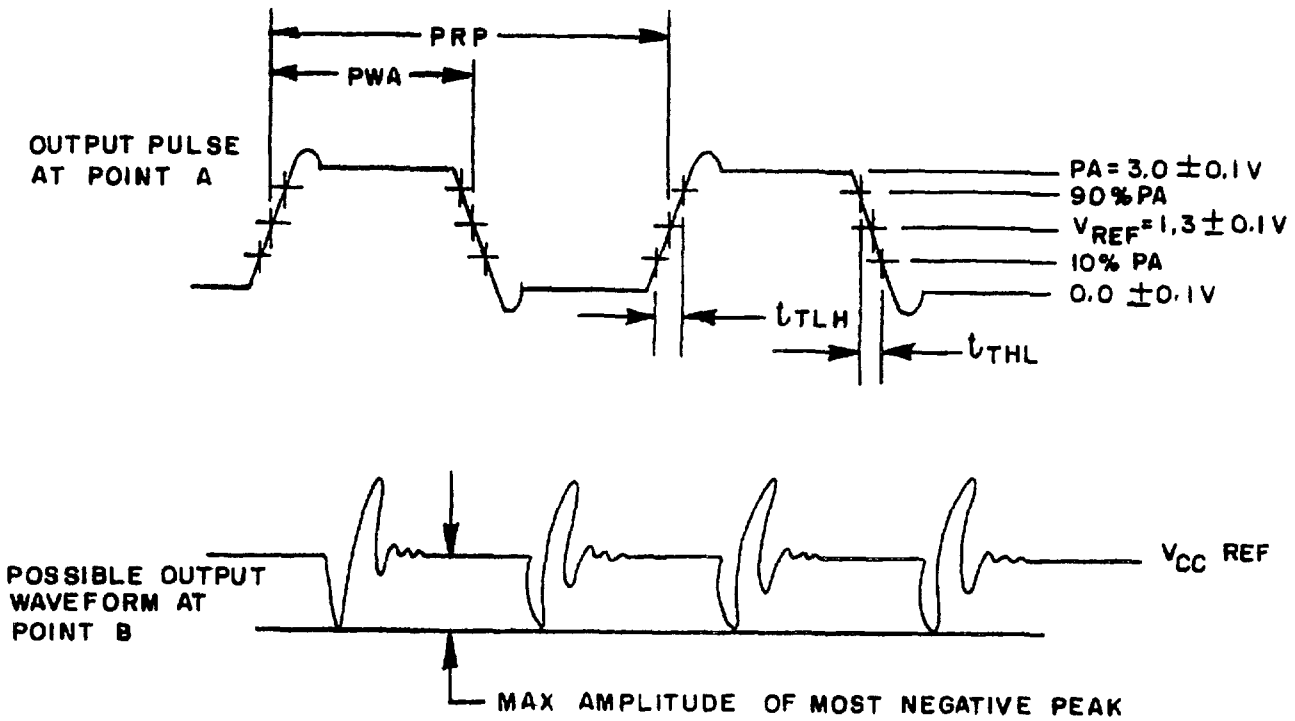
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 25. Power supply transient amplitude test circuit for table XXI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
PWA = 200 ±20 ns.
PRP = 400 ±40 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns.

FIGURE 26. Power supply transient amplitude test circuit waveforms for table XXI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XXII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Z	Power supply filter capacitance	1	Open	1.54	2.86	µF	27

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XXIII. Input capacitance test.

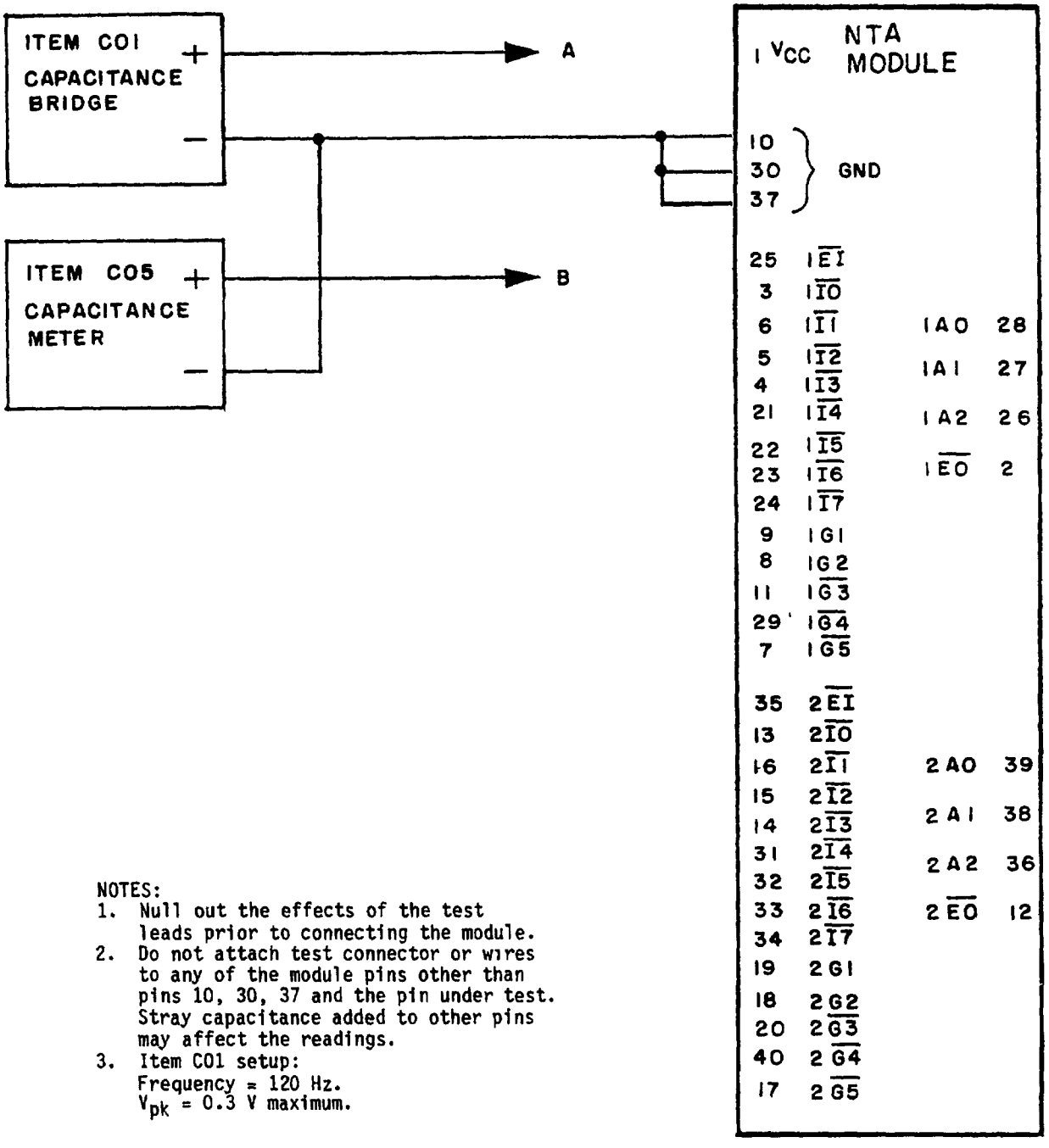
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA1	$1\overline{EI}$ Input capacitance	Open	25	2	15	pF	27
AA2	$1\overline{IO}$ "	"	3	"	"	"	"
AA3	1G1 "	"	9	"	"	"	"
AA4	1G2 "	"	8	"	"	"	"
AA5	$1\overline{G3}$ "	"	11	"	"	"	"
AA6	$1\overline{G4}$ "	"	29	"	"	"	"
AA7	$1\overline{G5}$ "	"	7	"	"	"	"
AA8	$2\overline{EI}$ "	"	35	"	"	"	"
AA9	$2\overline{IO}$ "	"	13	"	"	"	"
AA10	2G1 "	"	19	"	"	"	"
AA11	2G2 "	"	18	"	"	"	"
AA12	$2\overline{G3}$ "	"	20	"	"	"	"
AA13	$2\overline{G4}$ "	"	40	"	"	"	"
AA14	$2\overline{G5}$ "	"	17	"	"	"	"
AA15	$1\overline{I1}$ "	"	6	"	20	"	"
AA16	$1\overline{I2}$ "	"	5	"	"	"	"
AA17	$1\overline{I3}$ "	"	4	"	"	"	"
AA18	$1\overline{I4}$ "	"	21	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TABLE XXIII. Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA19	1T5 Input capacitance	Open	22	2	20	pF	27
AA20	1T6 "	"	23	"	"	"	"
AA21	1T7 "	"	24	"	"	"	"
AA22	2T1 "	"	16	"	"	"	"
AA23	2T2 "	"	15	"	"	"	"
AA24	2T3 "	"	14	"	"	"	"
AA25	2T4 "	"	37	"	"	"	"
AA26	2T5 "	"	32	"	"	"	"
AA27	2T6 "	"	33	"	"	"	"
AA28	2T7 "	"	34	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.



NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pins 10, 30, 37 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item CO1 setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

FIGURE 27. Power supply filter and input capacitance test circuit for tables XXII and XXIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

TRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in mitting comments and suggestions for improvements. All users of military standardization documents are invited to provide gestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and led. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the blems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an nowledgement will be mailed to you within 30 days to let you know that your comments were received and are being sidered.

TE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of cification requirements on current contracts Comments submitted on this form do not constitute or imply authorization waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

EPARTMENT OF THE NAVY

FICIAL BUSINESS
NALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Electronic Systems Command
Washington, DC 20363

Attn: ELEX 8111

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:53Z
Check the source to verify that this is the current version before use.